UE16EC335 Control Systems.

INTRODUCTION

CLASS 1

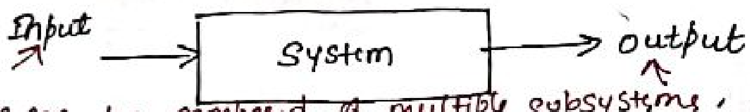
Drugs action

System: (Electrical system, Mechanical System, Chemical system, Biological system etc)

→ combination of components that act together to perform a function that is not possible with any of the indivisual.

No control action

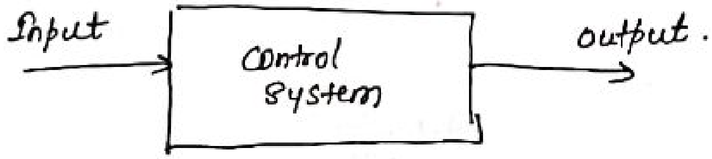
BULB LIGHT ON, OFF



(Aristotle: whole is greater than the sum of its parts)

Systems can be composed of multiple subsystems.

→ control system is a system, which provides the desired response by controlling the output.



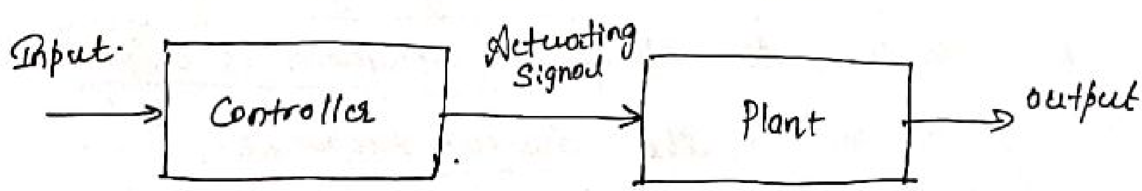
can you think for a while & give me some examples of control sytems that we use in real life.

Eg: car, ~~airconditioner~~ airconditioner, washing machine, valve in the tap to control water flow, microwave oven, (1 min full pow, 2 min half pow.) fan regulator, Gas Stove regulator, Refrigerator, light source with variable illumination, Robotic surgery, aircraft, Flush tank in toilet (water level is maintained) water level control system. gas geyser, Mixers (speed is controlled), Breaks, Bicycle → Balancing, steering (control is manual here), lift (elevator), clock.

Control system → Open loop control system

Control system → Closed loop contool system.

Open loop control system

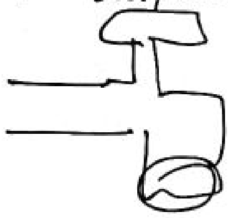


In open loop control system, output is not fed-back to the input, so control action is independent of the desired output.

Here an input is applied to a controller & it produces an actu Signal or controlling Signal. This signal is given as an input to a plant or process which is to be controlled. So the plant produces an output, which is controlled.

open loop eg: Bread toaster

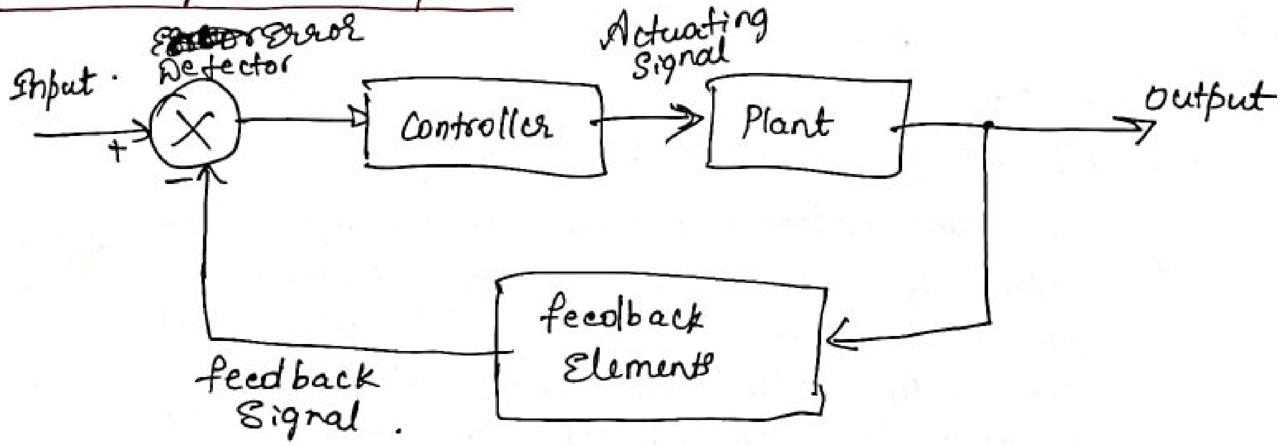
Eg:



tap. without valve
no control of water flow.

Tap with valve

Closed loop control system



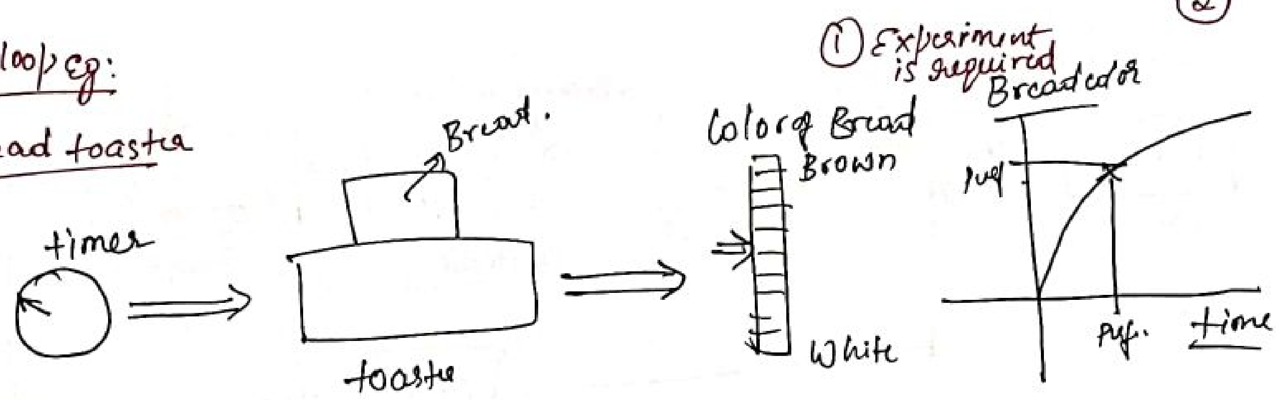
The error detector produces an error signal, which is the difference b/w the input and the feed back signal, This feedback signal is obtained from the block of feedback elements, by considering the o/p of system as an input to this block. Instead of the direct i/p, the error signal is applied as an input to a controller.

controller produces an actuating signal which controls the plant. In this, the o/p of control system is adjusted automatically till we get the desired response.

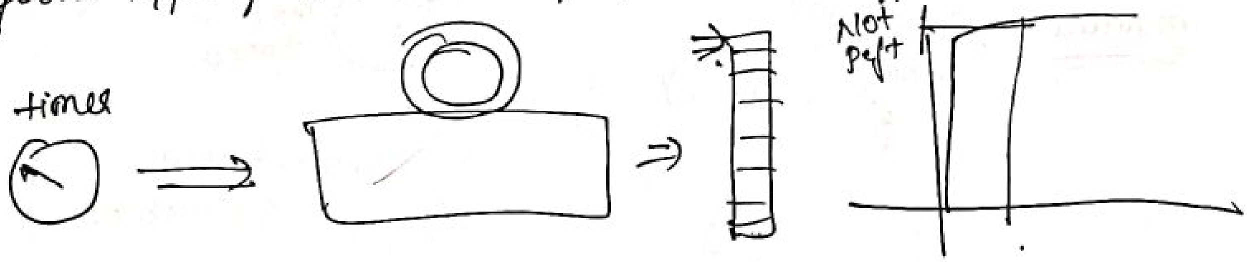
Hence closed loop systems are automatic control system

Open loop eg:

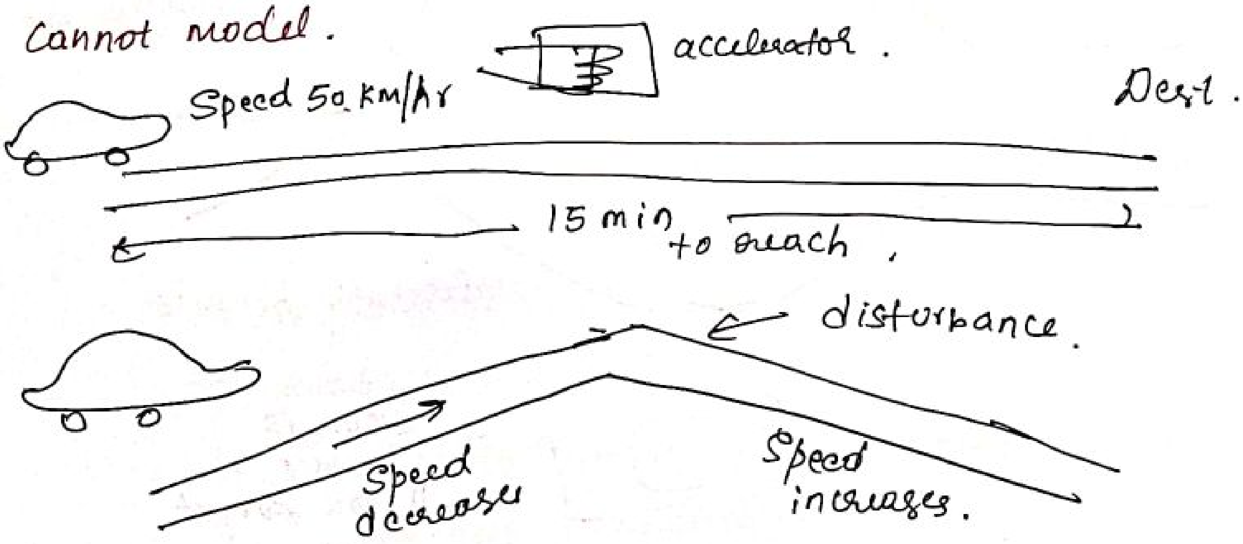
Bread toaster



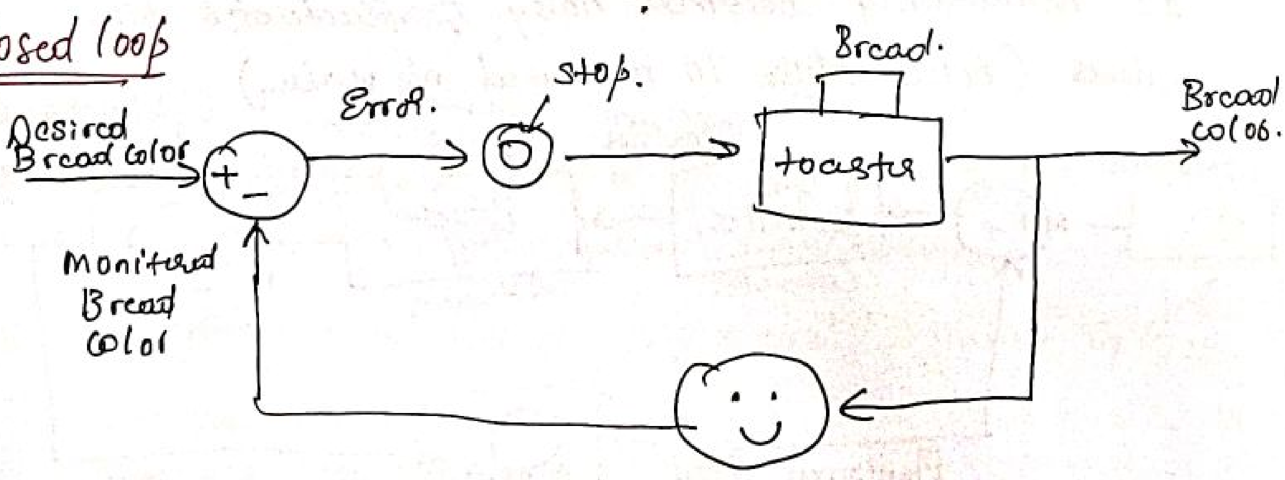
Different type of Bread then graph will be different.

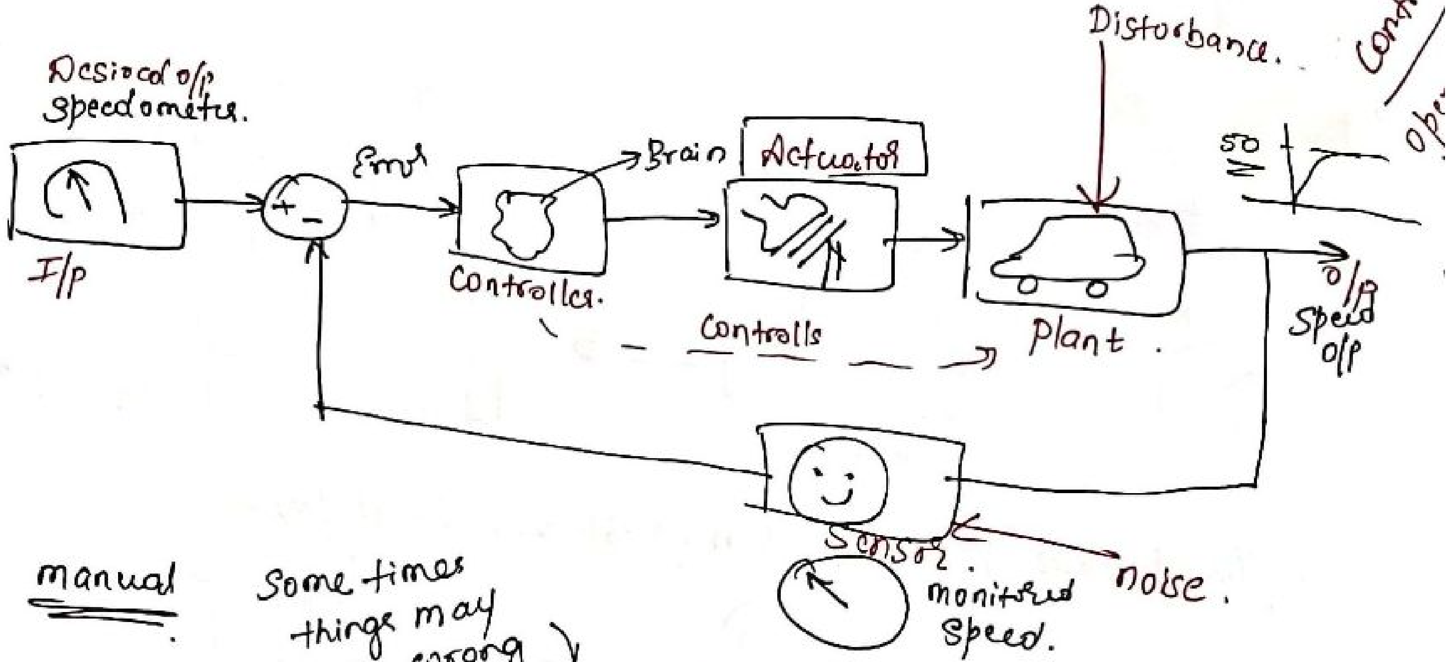


② One more Problem with open loop is if there is disturbance it cannot model.



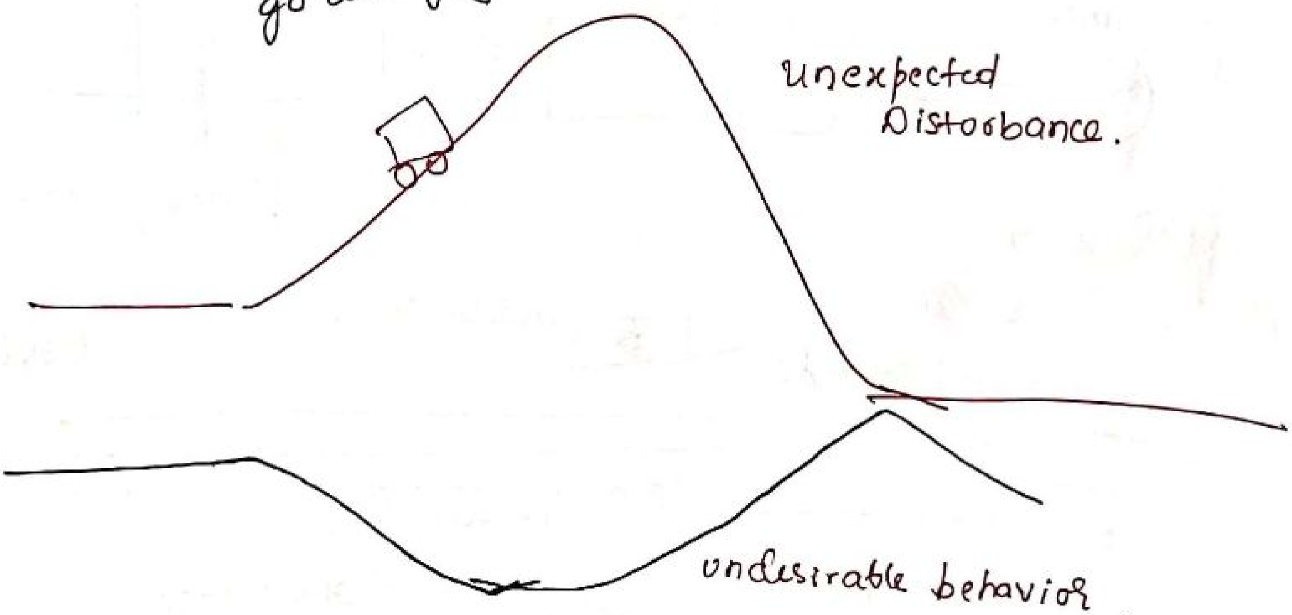
Closed loop

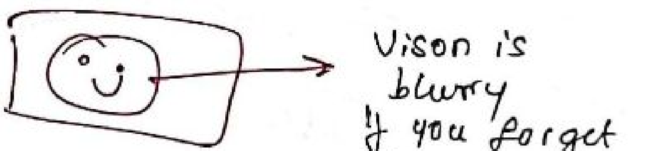




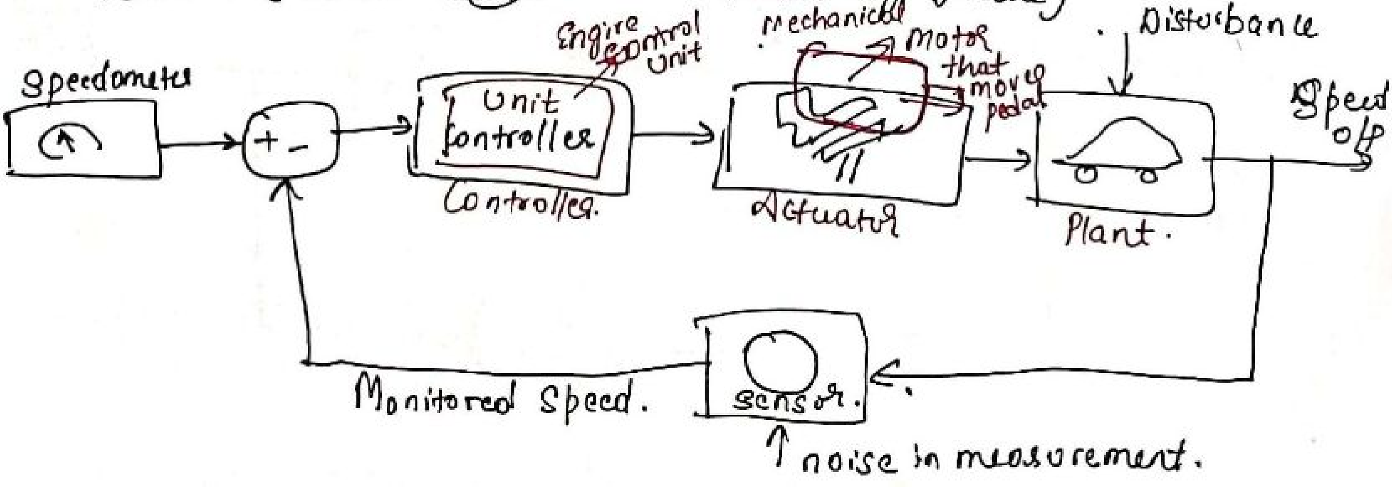
manual

Some times things may go wrong





The monitoring becomes noisy & fluctuates b/w values (noise ~~state~~ in measured o/p value)



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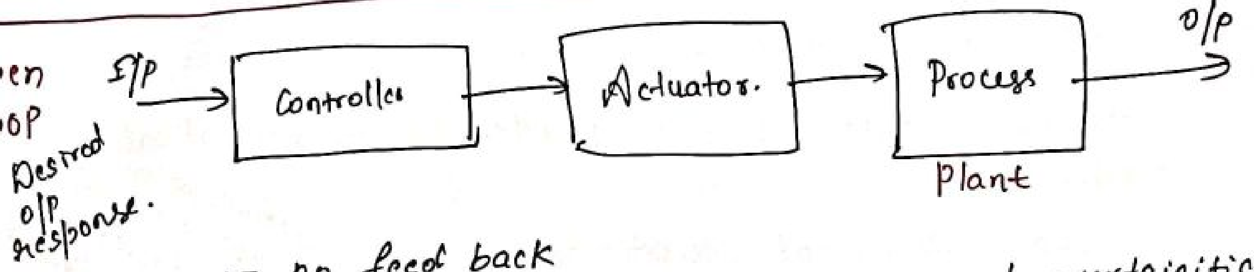
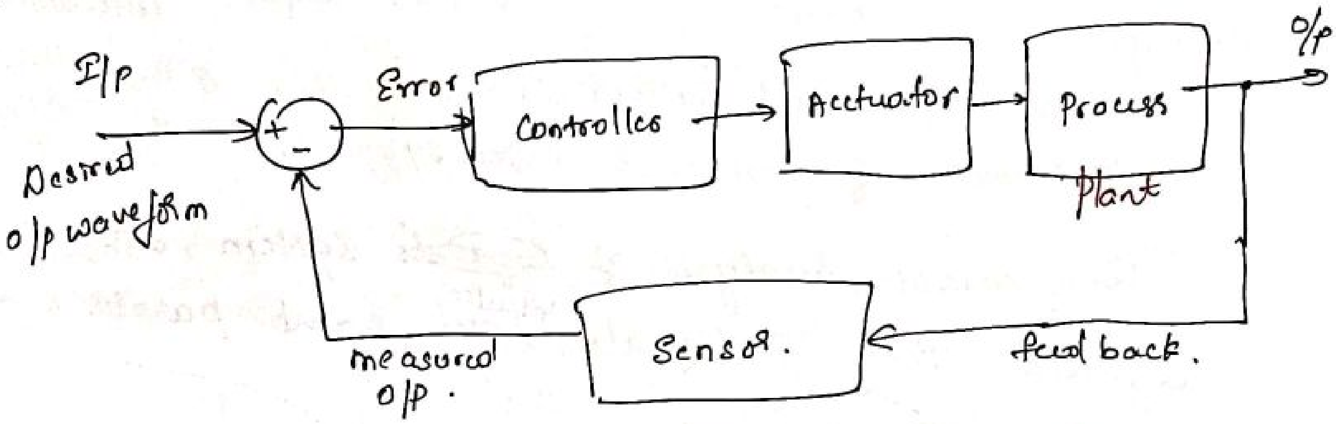
Control System Configuration

Open Loop



- no feed back
- o/p is effected due to disturbance / uncertainities.
- pre-set operation
- stability is not ensured / guaranteed.

Closed loop



- Negative feedback.
- Stability can be assured.
- plant / process can be made insensitive to disturbances & uncertainities.
- complexity of the system.

Control System Design

① Establishment of goals / objectives, definition of variables to be controlled (specification identification)

② Mathematical modelling : Differential / Difference Eq n.

③ controller design.

④ Software simulation & Analysis.

⑤ Hardware implementation.

Lesson plan

Unit 4

Unit 1: Mathematical models of system.

System behavior is represented by mathematical model.

→ Differential Equations.

→ Transfer functions

→ State space representations. (Representation of dynamic behavior of physical system using state variable) nth order diff eqn ÷ in n 1st order diff eqn.

→ Blockdiagram, Signal flow graph (system consists of lot of subsystems that are interconnected).

SISO
MIMO
Systems

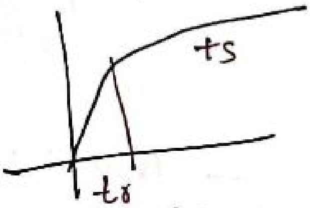
Timedomain Analysis — Frequency domain analysis.

Unit 2:

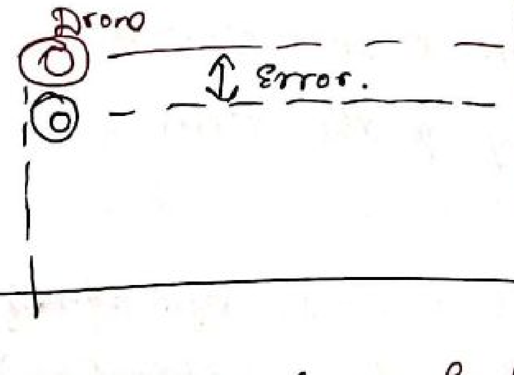
Feed back control system characteristics & Performance of feed back control system.

Time domain analysis of ~~signals~~ system, for different test i/p signals. impulse, step, ramp, parabola etc.

Transient response — Steady state respons



Steady state Errors.



→ Aware of the role of error signal on CS,
→ Understant the effect of feed back on system parameters, transient & steady state response

Effect of pole location on 2nd order system & transient response

Unit 3

The stability of linear feed back systems and root locus method (relative stability). (Eg salt range for particular dish)

range of K for which system is stable

→ Routh-Hurwitz Stability Criteria.

→ root locus concept & procedure
Learn sketching root locus & its role in CS.

Diff b/w absolute stability relative stability

Unit 4

Frequency response methods and stability in frequency domain.

→ frequency domain specification of system

→ Bode plots, How do we construct Bode plots.

→ polar plots., Rules for drawing polar plots

↳ Nyquist plot (for stability analysis)

Unit 5

Design of feedback control systems & Design of state variable feed back systems.

(in series with plant) → Cascade compensator. Compensators.

→ lag compensator

→ lead compensator.

→ lag-lead compensator.

→ lead-lag compensator

used to stabilize the unstable system.

feedback compensator. (feed back with plant)

→ Controllers

→ proportional controller → produces o/p proportional to error signal.

produces o/p, which is derivative of error signal. → Derivative controller.

→ Integral controller → produces o/p, which is integral of error signal.

→ Different combinations

PD, PI, PID.

P.C → used to change Transient response as per requirement

PD.C → used to make unstable CS into stable one

PIC → used to decrease steady state error

PID → used to improve stability & decrease the steady state error.

Text Book

① T1:- "Modern Control Systems", RC Dorf & RH. Bishop. 11th edition, Pearson education 2009.

② "control system Engineering", I.J. Nagrath, M Gopal 5th edition New age international Publications 2007.

③ "Modern control Engineering", K. Ogata, 5th edition. Pearson education Asia, 2010.

Project based ⇒ matlab/scilab programs.

ISA 1 — 20

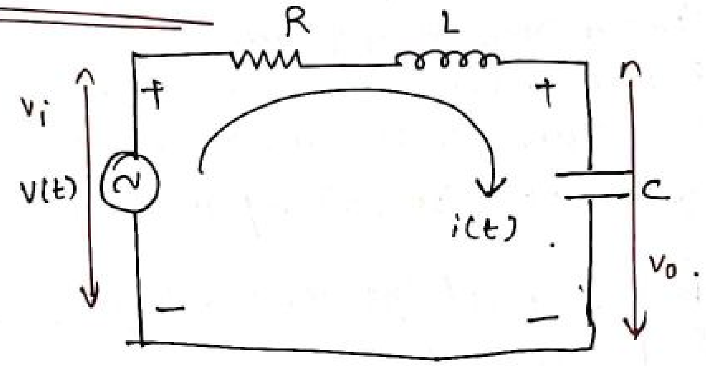
ISA 2 — 20.

Proj → 20

ESA → 40.

Mathematical Model of Electrical System

Simple System



Kirchoff's voltage law.

$$V(t) = R\cdot i(t) + L\cdot \frac{d\,i(t)}{dt} + \frac{1}{C}\cdot\int_{-\infty}^{t} i(t)\cdot dt \longrightarrow (1)$$

Integro-differential Eqn.

differentiate again to get 2nd order DE

$$\boxed{\frac{dv(t)}{dt} = R\cdot\frac{d\,i(t)}{dt} + L\cdot\frac{d^2 i(t)}{dt} + \frac{1}{C}\cdot i(t)}$$

take laplace transform.

$$S\cdot V(S) = R\cdot S\,I(S) + L\cdot S^2\,I(S) + \frac{1}{C}I(S)$$

$$C\,S\cdot V(S) = CRS\,I(S) + LCS^2\,I(S) + I(S)$$

$$CS\cdot V(S) = I(S)(1 + RCS + LCS^2)$$

O/P

$$\boxed{\frac{I(S)}{V(S)} = \frac{CS}{LCS^2 + RCS + 1}}$$

I/P.

Depending on the signal of interest we get different transfer function.

$$\boxed{\frac{V_C(S)}{V(S)} = \frac{1}{LCS^2 + RCS + 1}}$$

from Eqn 1

$$\boxed{V(t) = R\,i(t) + L\cdot\frac{d\,i(t)}{dt} + V_c(t)} \longrightarrow (2)$$

~~take~~

$$i = \frac{C\cdot d\,V_c(t)}{dt} \longleftarrow \therefore V_c(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)\cdot d\tau$$

$$\Rightarrow V(t) = R\cdot C\cdot\frac{d\,V_c(t)}{dt} + L\cdot C\cdot\frac{d^2 V_c(t)}{dt} + V_c(t)$$

take laplace transfrom.

$$V(s) = R\cdot C\cdot s\,V_c(s) + LC\,s^2 V_c(s) + V_c(s)$$

$$\boxed{\frac{V_c(s)}{V(s)} = \frac{1}{LCs^2 + RCs + 1}}$$

O/P: $V_c(s)$, I/P: $V(s)$

$$V_c = \frac{Q}{C} \qquad Q = \int_{-\infty}^{t} i(\tau)\cdot d\tau \Rightarrow i(t) = \frac{dQ}{dt}$$

$$I(s) = s\,Q(s)$$

$$Q(s) = \frac{I(s)}{s}$$

~~from Eqn (2)~~

$$\cancel{V(t) = R\cdot\frac{d\,Q(t)}{dt} + L\cdot\frac{d^2 Q(t)}{dt^2} + \frac{Q(t)}{C}}$$

~~take laplace.~~

$$\cancel{V(s) = R\cdot s\cdot Q(s) + L\cdot s^2 Q(s) + \frac{1}{C}\cdot Q(s)}$$

WKT

$$\frac{I(s)}{V(s)} = \frac{Cs}{LCs^2 + RCs + 1}$$

$$\frac{\cancel{s}\cdot Q(s)}{V(s)} = \frac{C\cancel{s}}{LCs^2 + RCs + 1}$$

$$\boxed{\frac{Q(s)}{V(s)} = \frac{C}{LCs^2 + RCs + 1}}$$

$$\frac{I(s)}{V(s)} = \frac{Cs}{LCs^2 + RCs + 1}$$

$$\frac{V_c(s)}{V(s)} = \frac{1}{LCs^2 + RCs + 1}.$$

$$\frac{Q(s)}{V(s)} = \frac{Cs^2}{LCs^2 + RCs + 1}$$

Denominator is similar,

⇒ it defines who & what about that system.

⇒ heart of the system.

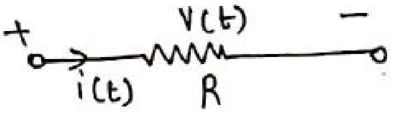
Next ⇒ Modelling of Mechanical Systems.

<u>Note</u>

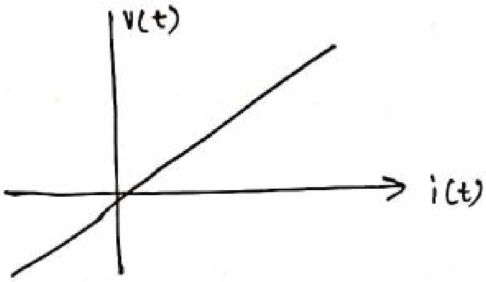
Energy stored in capacitor $= \frac{1}{2}CV^2$

" " " inductor $= \frac{1}{2}Li^2$

Electrical System:



$V(t) = R\,i(t)$



⇒ ohms law is true under idealizing assumptions.

not every device will follow this curve.

Eg: Tunnel diode. (Many things are non-linear).

Assumptions ~~under the~~

① Linearity: system that satisfy superposition theorem.

→ additivity
→ homogenity.

$$\alpha_1 u_1 + \alpha_2 u_2 \xleftrightarrow{T} \alpha_1 y_1 + \alpha_2 y_2$$

② Time invariant: If $u(t)$ is the i/p to the system and the corresponding o/p is $y(t)$, then the system is said to be time invariant if $u(t-\tau)$ gives the o/p $y(t-\tau)$, $\tau > 0$

But many physical systems are non-linear & time varying, But for analysis purpose we make the assumptions that system are linear and time invariant. we try to find the linear approximations of non linear systems.

③ lumped parameter: The physical size of the element is negligibly small when compared to the wavelength of the electromagnetic wave propogation.

Eg: R, C, L etc.

$V = \lambda f$ where V = velocity, λ = wavelength f = frequency

WKT EM wave in vaccum has same velocity as light.

$V = 3\times10^8$ m/s.

$f = 50\,HZ$ to $60\,HZ$

then $\lambda = 5\times10^6 = 5000\,km$

which can be very large comparable with physical size of elements like RLC, as a result physical size looks like a point wrt λ value.

Why do we make LTI assumptions?

> We can get ordinary differential equations.

> If the elements are time varying, then we get partial Differential Equations.

Note Distributed element: The physical size of element is comparable to the wavelength of electromagnetic wave propogation.

Eg: Transmission line ⇒ 1500 km.

KCL, KVL are applicable only when the elements are lumped

LAPLACE TRANSFORMS

$s = \sigma + j\omega$

Definition $\mathcal{L}\{f(t)\} = \int_{-\infty}^{\infty} f(t)\, e^{-st}\, dt$ ⇒ Bilateral.

$\mathcal{L}\{f(t)\} = \int_{0^-}^{\infty} f(t)\, e^{-st}\, dt$ ⇒ unilateral.

(some books will have 0^+, which is incorrect)

Causal system ⇒ system depends upon past and present i/p's but not future i/p's.

(If in speech/image processing we have given complete signal then we can use Bilateral).

Existance of Laplace transform (e^{t^2} grows too fast as $t \to \infty$ doesnot have Laplace transform)

① $f(t)$ must be piecewise continuous (ie. finite no of discontinuities over finite interval of bound)

② For $\int_0^{\infty} |f(t)|\, e^{-\sigma t}\, dt < \infty$ as $e^{-\sigma t} f(t) \to 0$ as $t \to \infty$.

(absolutely integrable)

The range

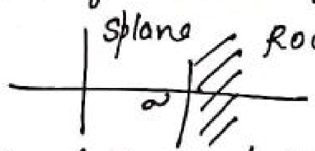
Properties of ROC

① ROC

The range of 'σ' for which LT converges is called ROC.

Properties of ROC

1. ROC of $F(s)$ includes strip lines $\parallel^{el}$ to $j\omega$ axis in s-plane
2. If $f(t)$ is absolutely Integrable & finite duration, then ROC is entire s-plane.
3. If $f(t)$ is Right sided signal, then ROC $Re\{s\} > \sigma$



4. If $f(t)$ is left sided, then ROC $Re\{s\} < \sigma$
5. If $f(t)$ is 2 sided signal, then ROC is combination of 2 regions.

Important transformpairs $(t \geq 0)$

$f(t)$ $(t \geq 0)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
t	$\frac{1}{s^2}$
e^{-at}	$\frac{1}{s+a}$
$\sin\omega t$	$\frac{\omega}{s^2+\omega^2}$
$\cos\omega t$	$\frac{s}{s^2+\omega^2}$
t^n	$\frac{n!}{s^{n+1}}$
$t \cdot e^{-at}$	$\frac{1}{(s+a)^2}$
$e^{-at}\sin\omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at}\cos\omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$

Note

$e^{-at} f(t) \longleftrightarrow F(s+a)$

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$$\mathcal{L}\{f^{n}(t)\} = \mathcal{L}\left\{\frac{d^{n}f(t)}{dt^{n}}\right\} = s^{n}F(s) - s^{n-1}f(0^-) - s^{n-2}f'(0^-) \cdots - f^{n-1}(0^-)$$

$$\mathcal{L}\left\{\int_0^t f(t)\cdot dt\right\} = \frac{F(s)}{s}$$

$$\mathcal{L}\left\{\int_{-\infty}^{t} f(t)\cdot dt\right\} = \frac{F(s)}{s} + \frac{1}{s}\int_{-\infty}^{0} f(t)\cdot dt.$$

Initial value theorem: $f(0^-) = \lim_{t\to 0^-} f(t) = \lim_{s\to\infty} sF(s)$

Final value theorem:

$$f(\infty) = \lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

Transfer function

$$G(s) = \frac{N(s)}{\Delta(s)} = \frac{b_m s^m + b_{m-1}s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}$$

Note: ① we always maintain co-efficient of highest degree in $\Delta(s)$ as 1.

② degree of Numerator polynomial is strictly less than ~~the~~ degree of denominator polynomial ($m < n$) (otherwise system will be noncausal). (strictly proper TF)

Transfer function of a linear system is defined as the ratio of L.T of output variable to the L.T of i/p variable with all initial conditions assumed to be zero.

— Defined for LTI system.

— Not valid for time varying system / Non linear system.

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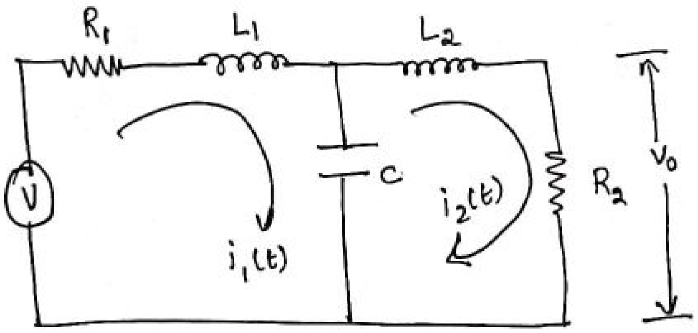
eg. (2)



loop (1)

$$V(t) = R_1 i_1(t) + L_1 \frac{d}{dt} i_1(t) + \frac{1}{C}\int_{-\infty}^{t} \left(i_1(\tau) - i_2(\tau)\right) d\tau \longrightarrow (a)$$

loop (2)

$$0 = L_2 \frac{d\, i_2(t)}{dt} + R_2 i_2(t) + \frac{1}{C}\int_{-\infty}^{t} \left(i_2(\tau) - i_1(\tau)\right) d\tau \longrightarrow (b)$$

where $V_0(t) = i_2(t) \cdot R_2 \longrightarrow$ (c)

under zero initial conditions apply LT on Bs of (a), (b), & (c)

$$(a) \Rightarrow V(s) = R_1 I_1(s) + L_1 s I_1(s) + \frac{1}{C}\left\{\frac{I_1(s) - I_2(s)}{s}\right\}.$$

$$(b) \Rightarrow 0 = L_2 s \cdot I_2(s) + R_2 I_2(s) + \frac{1}{C}\left\{\frac{I_2(s) - I_1(s)}{s}\right\}$$

$$(c) \Rightarrow V_0(s) = I_2(s) \cdot R_2 \Rightarrow \boxed{I_2(s) = \frac{V_0(s)}{R}}$$

$$(a) \Rightarrow V(s) = \left[R_1 + L_1 s + \frac{1}{Cs}\right] I_1(s) - \frac{1}{Cs} I_2(s)$$

$$(b) \Rightarrow \left[\frac{1}{Cs} + L_2 s + R_2\right] I_2(s) = \frac{1}{Cs} I_1(s)$$

$$\Rightarrow I_1(s) = Cs\left[\frac{1}{Cs} + L_2 s + R_2\right] I_2(s)$$

$$\boxed{I_1(s) = \left[1 + L_2 C s^2 + R_2 C s\right] I_2(s)}$$

Now substitute in V(s) Eqn.

$$V(s) = \underset{\frac{V_o(s)}{R_2}}{I_2(s)}\left[R_1 + L_1 s + \frac{1}{cs}\right]\left[1 + L_2 c s^2 + R_2 c s\right] - \frac{1}{cs}\left\{\frac{V_o(s)}{R_2}\right\}.$$

$$V(s) = \frac{1}{cs}\left[\left[R_1 cs + L_1 c s^2 + 1\right]\left[1 + L_2 c s^2 + R_2 cs\right] - 1\right]\frac{V_o(s)}{R_2}.$$

$$\frac{V_o(s)}{V(s)} = \frac{R_2 cs}{(R_1 cs + L_1 cs^2 + 1)(1 + L_2 cs^2 + R_2 cs) - 1}$$

$$D(s) = R_1 cs + L_1 c s^2 + \cancel{1} + R_1 c^2 s^3 L_2 + L_1 L_2 c^2 s^4 + L_2 c s^2 + R_1 R_2 c^2 s^2 + L_1 R_2 c^2 s^3 + R_2 cs - \cancel{1}$$

$$D(s) = cs\left\{R_1 + L_1 s + R_1 L_2 cs^2 + L_1 L_2 c s^3 + L_2 s + R_1 R_2 cs + L_1 R_2 cs^2 + R_2\right\}.$$

$$D(s) = cs\left\{(L_1 s + R_1) + (L_2 s + R_2) + L_1 L_2 c s^3 + (R_1 L_2 + L_1 R_2) cs^2 + R_1 R_2 cs\right\}.$$

$$D(s) = cs\left\{L_1 L_2 cs^3 + (R_1 L_2 + L_1 R_2) cs^2 + (L_1 + L_2 + R_1 R_2 c)s + (R_1 + R_2)\right\}$$

$$\boxed{\frac{V_o(s)}{V(s)} = \frac{R_2}{L_1 L_2 cs^3 + (R_1 L_2 + L_1 R_2) cs^2 + (L_1 + L_2 + R_1 R_2 c)s + (R_1 + R_2)}}$$

* Since degree of $D(s)$ is 3, & also there are 3 Storage elements. Hence the order of system is $\underline{\underline{3}}$

* It has 3 poles and no finite zeros.

CLASS2

Modelling of Mechanical Systems

There are two types of mechanical systems based on the type of motion.

1. Translational mechanical systems.
2. Rotational mechanical systems.

Modelling of Translational Mechanical Systems.

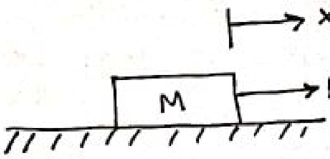
Translational mechanical systems move along a straight line. These systems mainly consist of 3 basic elements

→ mass
→ spring.
→ dashpot or damper.

If a force is applied to translational mechanical system, then it is opposed by opposing forces due to mass, elasticity and friction of the system. Since the applied force & the opposing forces are in opposite directions, the algebraic sum of the forces acting on the system is zero.

Mass: Mass stores kinetic energy

If a force is applied on a body having mass M, then it is opposed by an opposing force due to mass. This opposing force is proportional to the accelaration of the body. Assume elasticity and friction are negligible.



$F_m \propto a$ Newtons 2nd law.

$$F_m = Ma = M \cdot \frac{d^2x}{dt^2}$$

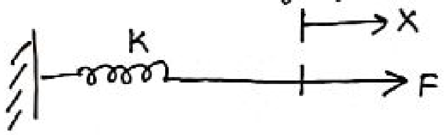
$$\boxed{F = F_M = M \cdot \frac{d^2x}{dt^2}}$$

where

F = applied force (N)
F_m = oppossing force due to mass
M = mass (Kg)
a = accelaration m/sec^2
x = displacement m

Spring: Spring stores potential energy

If a force is applied on spring K, then it is opposed by an opposing force due to elasticity of spring. This opposing force is proportional to the displacement of the spring. Assume mass & friction are negligible.



$$F_K \propto x$$

$$\Rightarrow F_K = K \cdot x.$$

$$\boxed{F = F_K = K \cdot x.}$$

where,

F = applied force. (N)

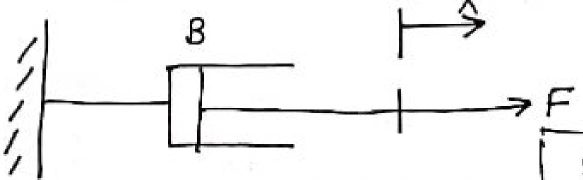
F_K = opposing force due to elasticity of spring

K = Spring constant (N/m) (elastisity or stifness)

x = displacement (m)

Dashpot or Damper (mechanical device, which resists motion via friction)

If a force is applied on Dashpot B, then it is opposed by an opposing force due to friction of the dashpot. This opposing force is proportional to the velocity of the body. Assume mass & elasticity are negligible.



$$F_b \propto v.$$

$$F_b = B \cdot v = B \cdot \frac{dx}{dt}$$

$$\boxed{F = F_b = B \cdot \frac{dx}{dt}}$$

where F = applied force (N)

F_b = Opposing force due to friction of dashpot.

B = frictional co-efficient (N·sec/m)

v = Velocity. (m/s)

x = displacement m

∴ Since degree ...

Rotational mechanical Systems move about a fixed axis. These Systems mainly consists of three basic elements

→ moment of inertia.

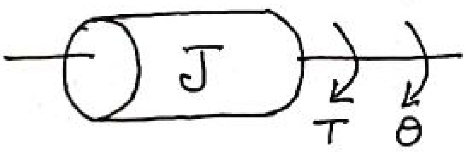
→ tortional spring.

→ dashpot.

If a torque is applied to a rotational mechanical system, then it is opposed by opposing torques due to moment of inertia, elasticity, and friction of the system. Since the applied torque & the opposing torques are in opposite directions, the algebraic sum of torques acting on the system is zero.

Moment of Inertia: In translational mechanical system, mass stores kinetic energy, similarly in rotational mechanical system, moment of inertia stores kinetic energy.

If a torque is applied on a body having moment of inertia J, then it is opposed by an opposing torque due to the moment of inertia. This opposing torque is proportional to angular accelaration of the body. Assume elasticity and friction are negligible.



$$T_j \propto \alpha$$

$$\Rightarrow T_j = J.\alpha = J.\frac{d^2\theta}{dt^2}$$

$$T = T_j = J.\frac{d^2\theta}{dt^2}$$

where T = applied torque. (Nm)

T_j ⇒ opposing torque due to moment of inertia.

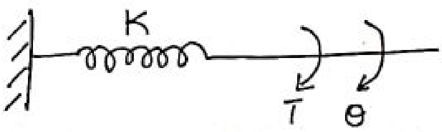
J ⇒ moment of inertia $Kg\text{-}m^2$

α ⇒ angular accelaration. rad/sec^2

θ ⇒ angular displacement. rad

Torsional Spring: In translational mechanical system, spring stores potential energy, III^{ly} in rotational mechanical system, torsional spring stores potential energy

If a torque is applied on torsional spring K, then it is opposed by an opposing torque due to the elasticity of the torsional spring. This opposing torque is proportional to the angular displacement of the torsional spring. Assume the moment of inertia and friction are negligible.



$$T_K \propto \theta$$
$$\Rightarrow T_K = K \cdot \theta$$
$$T = T_K = K \cdot \theta \, .$$

Where.

T = applied torque (N-m)

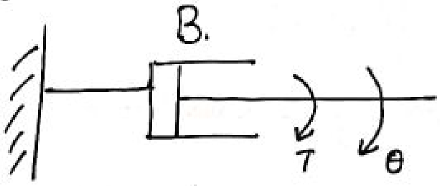
T_K = opposing torque due to elasticity of tortional spring

K = tortional spring constant (N-m/rad) (elasticity/stifness)

θ = angular displacement (rad)

Dashpot:

If a torque is applied on dashpot B, then it is opposed by an opposing torque due to the rotational friction of the dashpot. This opposing torque is proportional to the angular velocity of the body. Assume the moment of inertia and elastisity are negligible.



$$T_b \propto \omega$$
$$T_b = B \cdot \omega = B \cdot \frac{d\theta}{dt} \, .$$
$$T = T_b = B \cdot \frac{d\theta}{dt} \, .$$

where

T_b = opposing torque due to the rotational friction of the dashpot (Nm)

B = rotational friction co-efficient (N-m/(rad/sec)

ω = angular velocity (rad/sec)

θ = angular displacement. (rad)

Two Systems are said to be analogous to each other if the following two conditions are satisified.

- → The two systems are physically different
- → Differential equation modelling of these two systems are same.

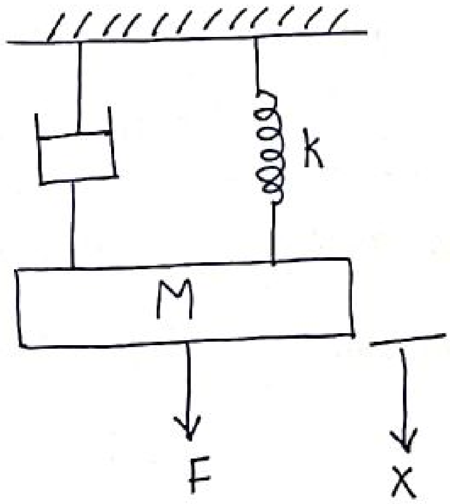
There are 2 types of electrical analogies of translational mechanical systems and rotational mechanical systems.

1. Force voltage analogy.
2. Force current analogy.

Force voltage Analogy:

In force voltage analogy, the mathematical Equations of translational mechanical system are compared with mesh equations of the electrical system

consider the following translational mechanical system.

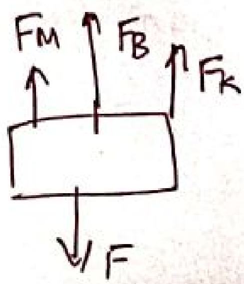


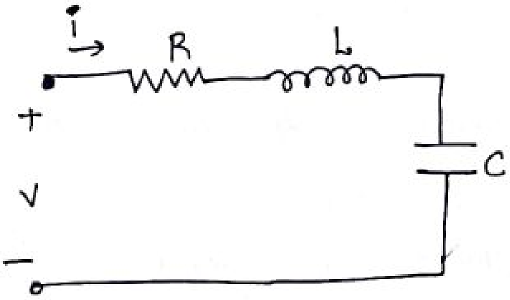
The force balanced Eq^n.

$$F = F_M + F_B + F_K.$$

$$\boxed{F = M.\frac{d^2x}{dt^2} + B.\frac{dx}{dt} + K.x} \rightarrow (1)$$

Consider the following electrical system as shown in below figure. (R, L, C in series)





Mesh Eqn. (KVL)

$$V = Ri + L\frac{di}{dt} + \frac{1}{C}\int i\,dt.$$

Substitute $i = \frac{dq}{dt}$

$$\boxed{V = R\frac{dq}{dt} + L\frac{d^2q}{dt^2} + \frac{q}{C}} \rightarrow (2)$$

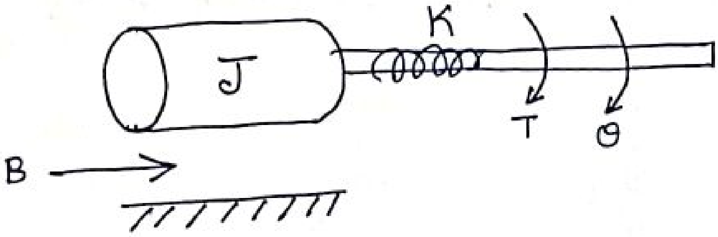
By comparing eq (1) & (2), we will get Force voltage analogy of translational system & Electrical system.

Translational Mechanical System	Electrical System
Force (F)	Voltage (V)
Mass (M)	Inductance (L)
Frictional Co-eff (B)	Resistance (R)
Spring constant (K)	Reciprocal of capacitance ($\frac{1}{C}$)
Displacement (x)	Charge (q)
velocity (v)	Current (i)

IIIly there is torque voltage analogy for rotational mechanical systems, we compare it with mesh equations of electrical system

(R2)

Consider the following Rotational mechanical system.



The torque balanced equation is.

$$T = T_j + T_b + T_K$$

$$\boxed{T = J.\frac{d^2\theta}{dt^2} + B.\frac{d\theta}{dt} + K\theta.} \rightarrow (3)$$

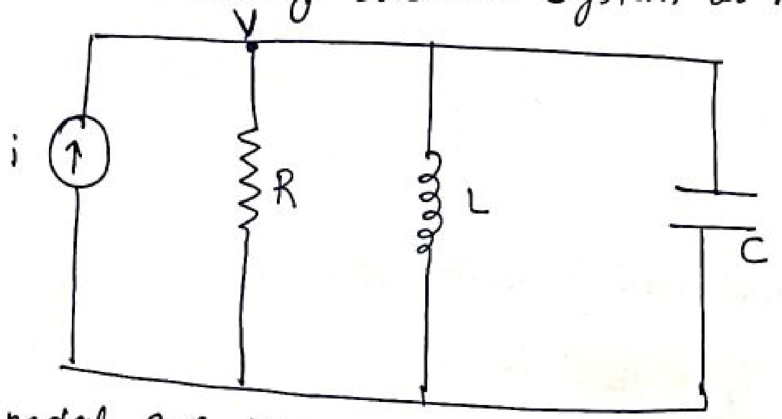
comparing Eqn (2) & (3), we will get analogous quantities of rotational mechanical system & electrical system.

Rotational Mechanical system	Electrical system.
Torque (T)	Voltage (V)
Moment of Inertia (J).	Inductance (L).
Rotational friction co-eff (B).	Resistance (R).
Tortional spring constant (K)	Reciprocal of capacitance ($\frac{1}{C}$)
Angular displacement (θ)	charge (q).
Angular velocity (ω).	current (i).

Force Current Analogy

In force current analogy, the mathematical equations of the translational mechanical system are compared with nodal equations of electrical system.

Consider the following electrical system as shown in fig.



The nodal Eqn is

$$i = \frac{v}{R} + \frac{1}{L}\int v \cdot dt + C \cdot \frac{dv}{dt}.$$

Substitute $\boxed{V = \frac{d\psi}{dt}}$

$$\boxed{i = C \cdot \frac{d^2\psi}{dt^2} + \left(\frac{1}{R}\right)\frac{d\psi}{dt} + \frac{\psi}{L}} \longrightarrow (4)$$

· By comparing Eqn (1) with (4), we get analogous quantities of the translational mechanical system and electrical system.

Translational Mechanical System	Electrical System.
Force (F)	current (i)
Mass (M)	Capacitance (C)
Frictional Co-eff (B)	Reciprocal of Resistance $(\frac{1}{R})$
Spring constant (K)	Reciprocal of Inductance $(\frac{1}{L})$
Displacement (x)	Magnetic flux (ψ)
Velocity (v)	Voltage (V)

$R_1+R_2)$

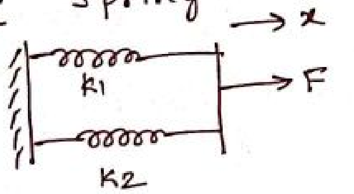
→ There is torque current analogy for rotational mechanical systems. (5)

IIIly. Mathematical Equations of the rotational mechanical system are compared with nodal mesh equations of the electrical system.

By comparing. Eqn (4) & (3), we will get the analogous quantities of the ~~translational~~ rotational mechanical system and electrical system.

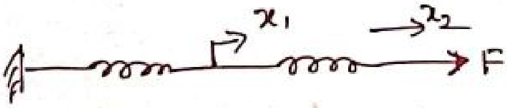
Rotational Mechanical system	Electrical system.
Torque (T)	Current (i)
Moment of inertia (J).	capacitance (C)
Rotational friction Co-eff (B)	Reciprocal of resistance ($1/R$)
Torsional spring constant (K)	Reciprocal of Inductance ($1/L$)
Angular displacement (θ)	Magnetic flux (ψ)
Angular velocity (ω)	Voltage (v)

Note: Spring



2 Springs has same displacement

$F = K_1 x + K_2 x = (K_1 + K_2) x$



2 Springs has different displacement but F is same

$K_1 x_1 = K_2 x_2 = F$

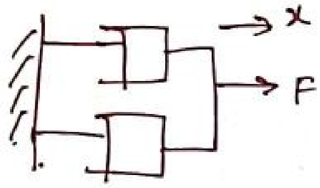
$x_1 = \frac{F}{K_1}$ $\quad x_2 = \frac{F}{K_2}$

$\frac{F}{K_{eq}} = \frac{F}{K_1} + \frac{F}{K_2} \Rightarrow \frac{1}{K_{eq}} = \frac{1}{K_1} + \frac{1}{K_2} = \frac{K_1 K_2}{K_1 + K_2}$

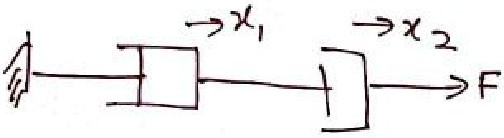
→ (4)

$$K_{eq} = \frac{K_1 K_2 \cdots K_n}{K_1 + K_2 + \cdots + K_n}$$

Damper / Dashpot



$$B_{eq} = B_1 + B_2$$

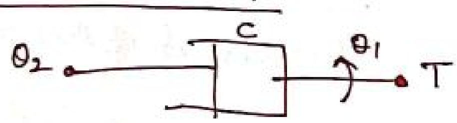


$$B_{eq} = \frac{B_1 B_2}{B_1 + B_2}$$

Rotational Spring

$$T = K(\theta_1 - \theta_2)$$

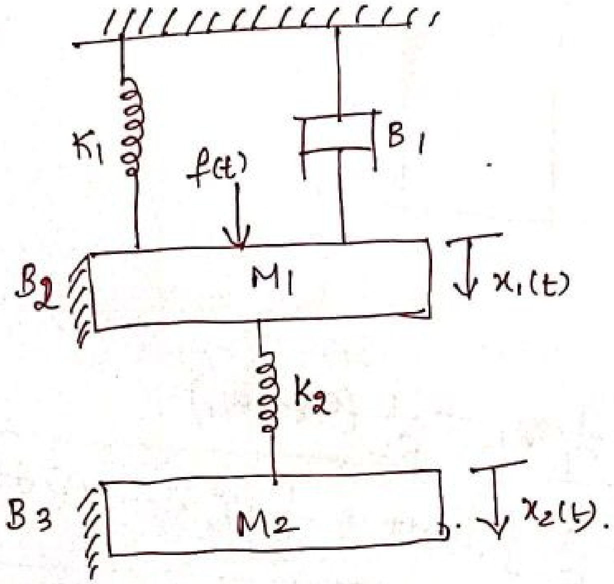
Rotational Damper.



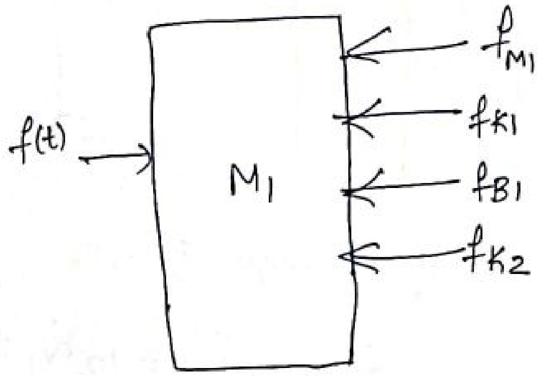
$$T = B(\dot{\theta}_1 - \dot{\theta}_2)$$

	F–V analogy	F–I analogy
F / T	V	I
M / J	L	C
B	R	$1/R$
K	$\frac{1}{C}$	$1/L$
x / θ	$\int_0^t i(\tau)\,d(\tau) = q$	$\psi \rightarrow \int_0^t v(\tau)\,dt$
v / ω. $\frac{dx}{dt}$	$= \dot{q} = i$ current.	v Voltage $= \frac{d\psi}{dt} = \dot{\psi}$
$\frac{dx^2}{dt^2} = a$	$\frac{di}{dt}$	$\frac{dv}{dt}$

1) For a mechanical system shown, draw the electrical N/W analogy based on force-current analogy. & Force-voltage analogy

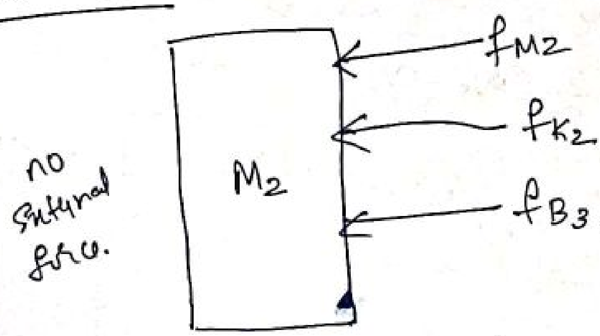


for M1



$$f(t) = M_1\ddot{x}_1 + B_1\dot{x}_1 + B_2\dot{x}_1 + K_1x_1 + K_2(x_1 - x_2) \rightarrow (1)$$

for M2



$$0 = M_2\ddot{x}_2 + B_3 \cdot \dot{x}_2 + K_2(x_2 - x_1). \longrightarrow (2).$$

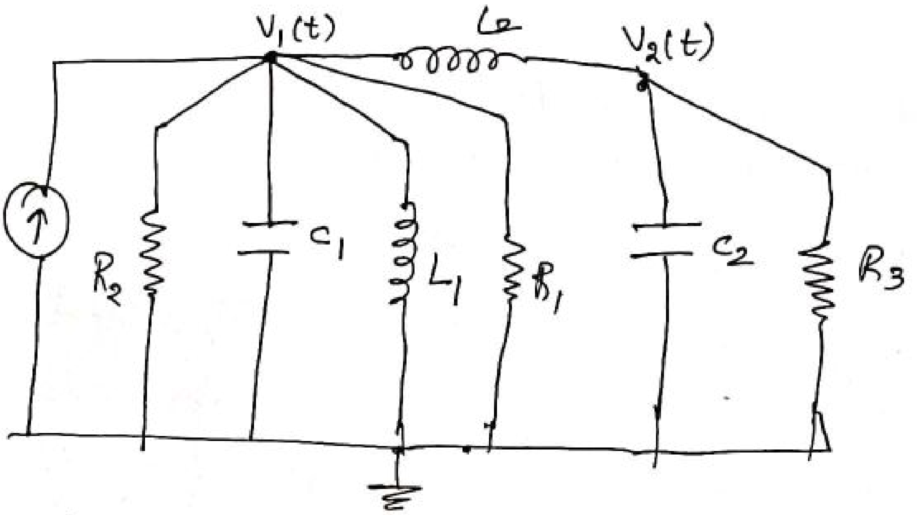
FI - analogy for Eqn (1)

$$I = C_1\ddot{\psi}_1 + \frac{1}{R_1}\cdot\dot{\psi}_1 + \frac{1}{R_2}\dot{\psi}_1 + \frac{1}{L_1}\psi_1 + \frac{1}{L_2}(\psi_1 - \psi_2)$$

$$\boxed{i(t) = C_1\frac{dv_1}{dt} + \frac{1}{R_1}v_1 + \frac{1}{R_2}v_1 + \frac{1}{L_1}\int_0^t v_1(\tau)d\tau + \frac{1}{L_2}\left(\int_0^t (v_1(\tau) - v_2(\tau))d\tau\right)} \rightarrow (3)$$

FI - analogy for Eqn (2)

$$\boxed{0 = C_2\cdot\frac{dv_2}{dt} + \frac{1}{R_3}\cdot v_2 + \frac{1}{L_2}\int_0^t (v_2(\tau) - v_1(\tau))d\tau = 0} \rightarrow (4)$$



FV – analogy for Eqn 1

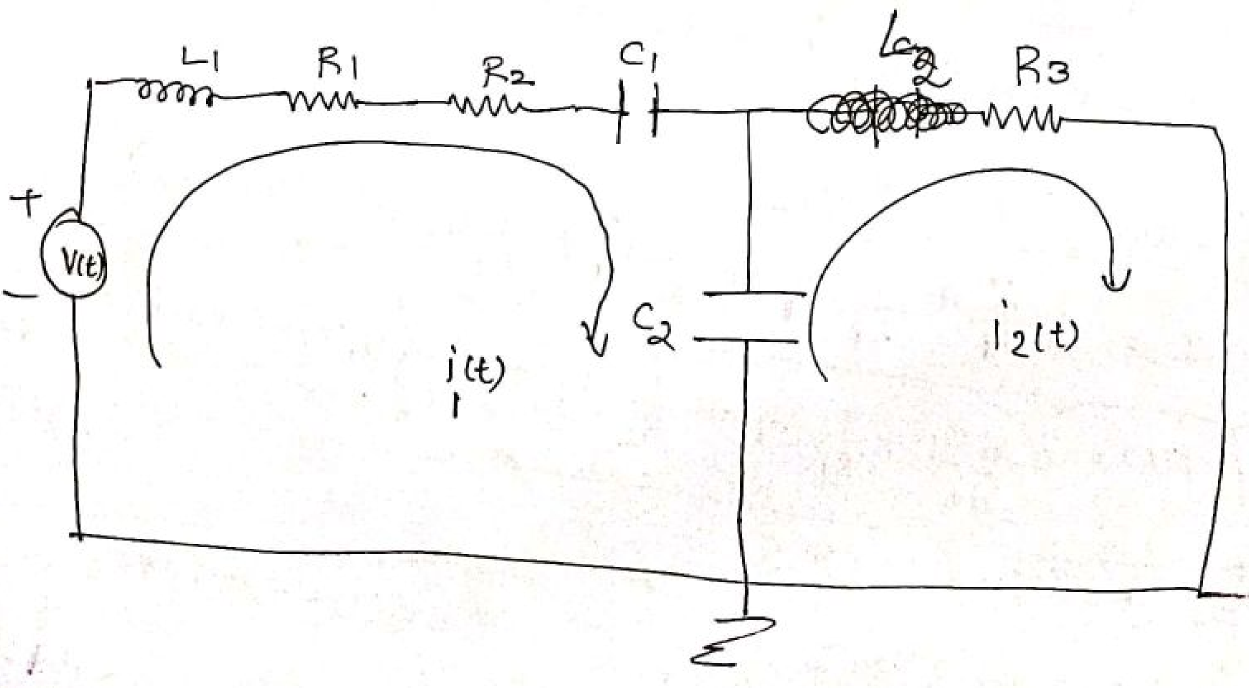
$$V(t) = L_1\ddot{q}_1 + R_1\dot{q}_1 + R_2\dot{q}_1 + \frac{1}{C_1}q_1 + \frac{1}{C_2}(q_2 - q_1)$$

$$\boxed{V(t) = L_1\frac{di_1}{dt} + R_1 \cdot i_1 + R_2 i_1 + \frac{1}{C_1}\int i_1(t)\,dt + \frac{1}{C_2}\int (i_2(t) - i_1(t))\,dt} \rightarrow (5)$$

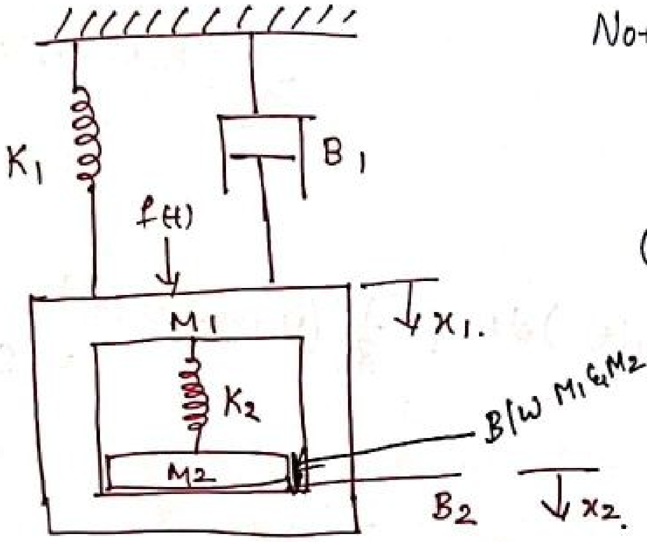
FV - analogy for Eqn 2

$$0 = L_2\ddot{q}_2 + R_3\dot{q}_2 + \frac{1}{C_2}(q_2 - q_1)$$

$$\boxed{0 = L_2\frac{di_2}{dt} + R_3 i_2 + \frac{1}{C_2}\int (i_2(t) - i_1(t))\,dt.} \rightarrow (6)$$



(2) For a mechanical system shown, draw electrical N/W based on FI analogy & FV analogy.



Note: (1) no of displacements must be equal to no of masses in the system

(2) no of displacements on either sides of a spring/damper must be different

Sol^n

FB diagram for M1

$f(t) \rightarrow M_1$

$f_{M_1} = M_1 \frac{d^2 x_1}{dt^2} = M_1 \frac{dv_1}{dt}$

$f_{K_1} = K_1 x_1 = K_1 \int v_1 dt$

$f_{B_1} = B_1 \dot{x}_1 = B_1 v_1$

$f_{K_2} = K_2 (x_1 - x_2) = K_2 \left[\int (v_1 - v_2) dt \right.$

$f_{B_2} = B_2 (\dot{x}_1 - \dot{x}_2)$

$$\therefore \boxed{f(t) = M_1 \frac{dv_1}{dt} + K_1 \int v_1(t)\, dt + B_1 v_1 + K_2 \int (v_1 - v_2) dt + B_2 (v_1 - v_2)} \rightarrow (1)$$

FB diagram for M2

$f_{M_2} = M_2 \cdot \frac{dv_2}{dt}$

$f_{K_2} = K_2 \int (v_2 - v_1) dt$

$f_{B_2} = B_2 (v_2 - v_1) = 0$

(no ext force) M_2

$$\boxed{M_2 \frac{dv_2}{dt} + K_2 \int (v_2 - v_1) dt + B_2 (v_2 - v_1) = 0} \rightarrow (2)$$

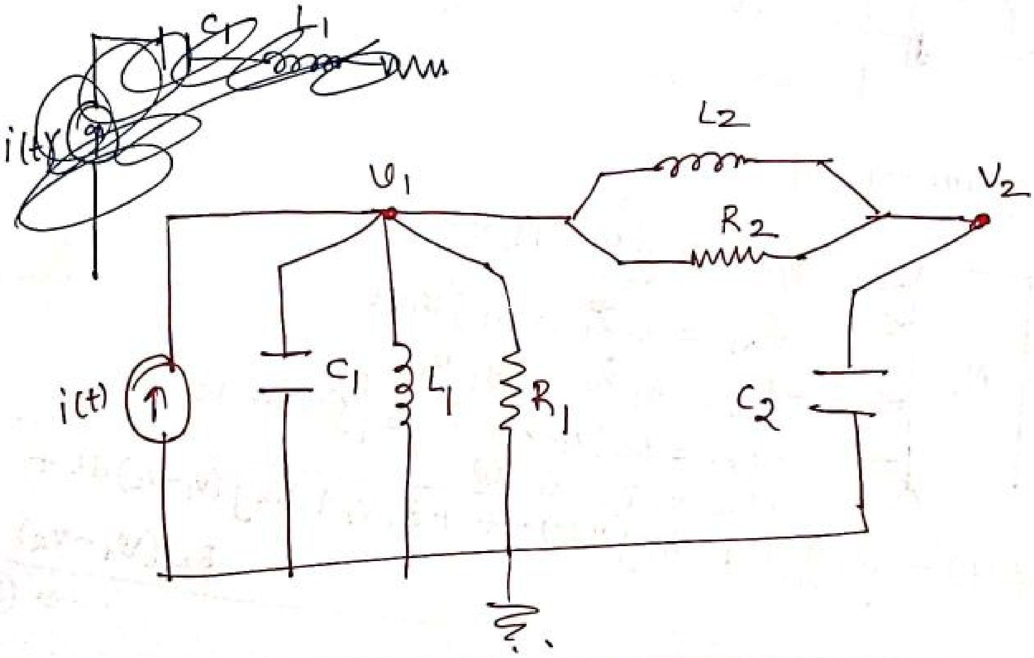
F I Analogy

Eqn ①

$$i(t) = C_1 \frac{dv_1}{dt} + \frac{1}{L_1}\int v_1(t)\,dt + \frac{v_1(t)}{R_1} + \frac{1}{L_2}\int \left(v_1(t) - v_2(t)\right)dt + \frac{1}{R_2}\left(v_1(t) - v_2(t)\right)$$

Eqn ②

$$C_2 \frac{dv_2}{dt} + \frac{1}{L_2}\int \left(v_2(t) - v_1(t)\right)dt + \frac{1}{R_2}\left(v_2(t) - v_1(t)\right) = 0.$$



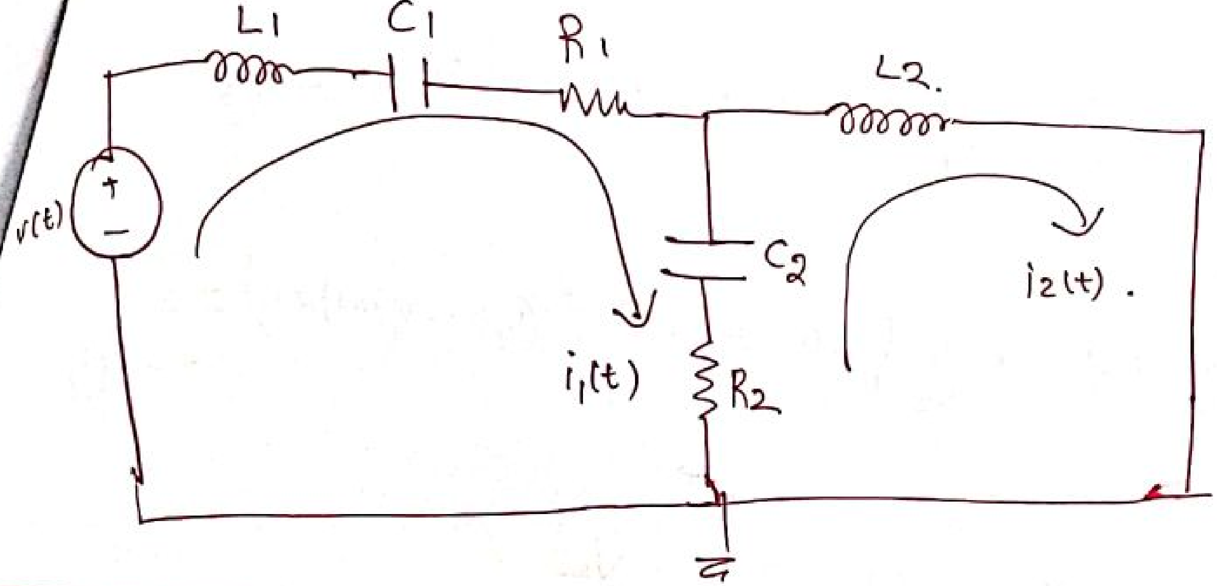
F V Analogy

Eqn ①

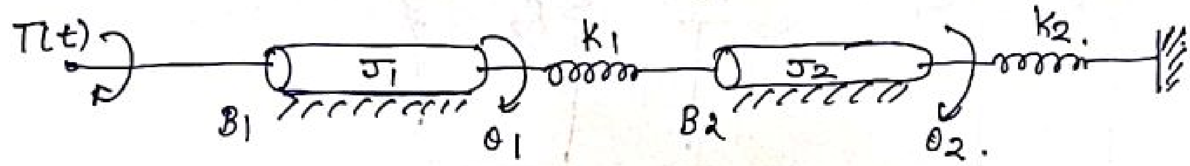
$$v(t) = L_1 \frac{di_1}{dt} + \frac{1}{C_1}\int i_1(t)\,dt + R_1\, i_1(t) + \frac{1}{C_2}\int \left(i_1(t) - i_2(t)\right)dt + R_2\left(i_1(t) - i_2(t)\right)$$

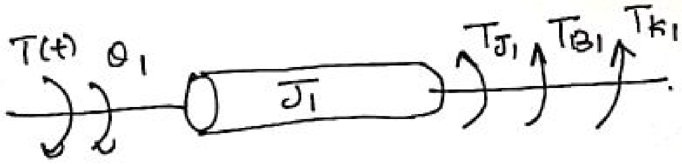
Eqn ②

$$L_2 \cdot \frac{di_2}{dt} + \frac{1}{C_2}\int \left(i_2(t) - i_1(t)\right)dt + R_2\left(i_2(t) - i_1(t)\right) = 0$$



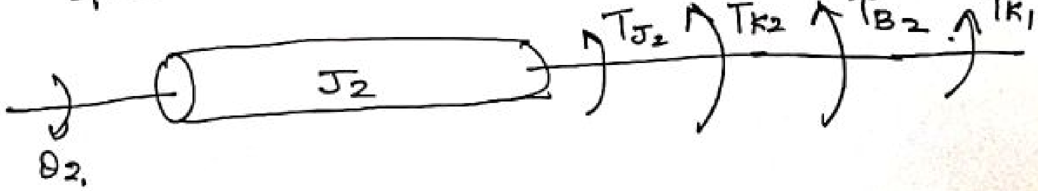
3) Draw electrical network based on torque – current analogy. Give all relevant equations.





$$J_1 \ddot{\theta}_1 + B_1 \dot{\theta}_1 + K_1(\theta_1 - \theta_2) = T$$

$$\boxed{J_1 \frac{d\omega_1}{dt} + B_1 \cdot \omega_1 + K_1 \left(\int (\omega_1(t) - \omega_2(t)) \, dt \right) = T} \rightarrow (1).$$



$$J_2 \ddot{\theta}_2 + B_2 \dot{\theta}_2 + K_2 \theta_2 + K_1(\theta_2 - \theta_1) = 0$$

$$\boxed{J_2 \frac{d\omega_2}{dt} + B_2 \omega_2 + K_2 \int \omega_2(t) \cdot dt + K_1 \int (\omega_2(t) - \omega_1(t)) \, dt = 0} \rightarrow (2).$$

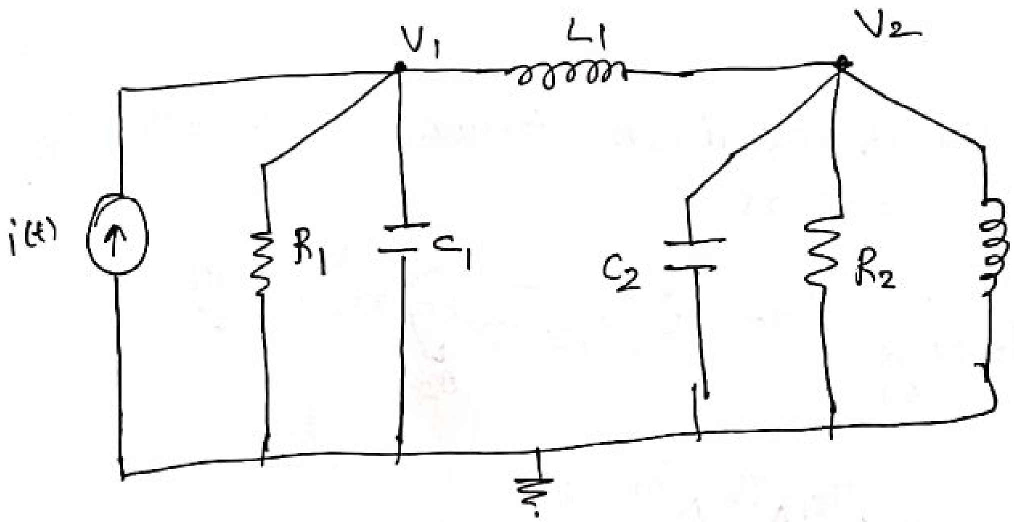
T-I-Analogy

from Eqⁿ (1).

$$C_1 \frac{dv_1}{dt} + \frac{v_1}{R_1} + \frac{1}{L_1}\int (v_1(t) - v_2(t))\, dt = i(t) \longrightarrow (3)$$

from Eqⁿ (2).

$$C_2 \cdot \frac{d v_2}{dt} + \frac{v_2}{R_2} + \frac{1}{L_2}\int v_2(t)\cdot dt + \frac{1}{L_1}\left(\int (v_2(t) - v_1(t))\,dt\right) = 0 \longrightarrow (4)$$



CLASS 3

1) Draw the Mechanical network:

- This is done by marking independent nodes that correspond to each displacement. Then, each element is connected between the nodes that correspond to the two displacements at each end of that element.
- The Mass & moment of Inertia are always connected from the reference node to the node representing its displacement.
- The Spring and dashpot elements are connected to the two nodes that represent the displacement of each end of the element.

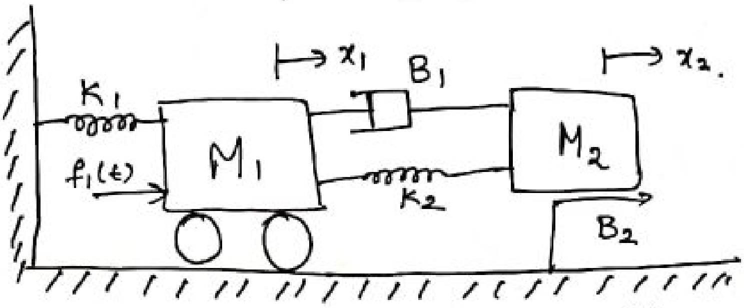
2) Once the mechanical network is drawn, the force/torque equation is written for each node by equating the sum of forces/torques at each node to zero. (This technique is similar to nodal analysis used in electrical circuits).

Checkpoints

① minimum number of displacements must be equal to the number of masses in the mechanical system.

② The displacements on either sides of spring/damper must be different.

Find the TF of the system shown below.

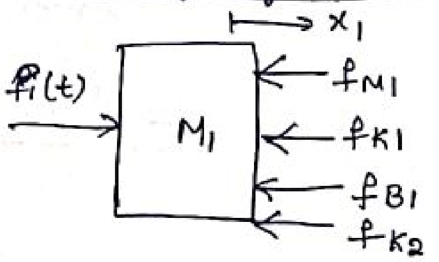


$I/p = F_1(s)$

$O/p \Rightarrow X_2(s)$

$$T.F = \frac{X_2(s)}{F_1(s)}$$

free body diagram for M1



$f_{M_1} = M_1 \ddot{x}_1$

$f_{B_1} = B_1(\dot{x}_1 - \dot{x}_2)$

$f_{K_1} = K_1 x_1$

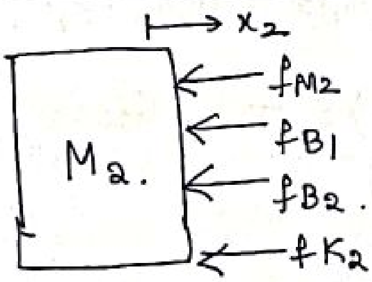
$f_{K_2} = K_2(x_1 - x_2)$

$$f_1(t) = M_1\ddot{x}_1 + B_1(\dot{x}_1 - \dot{x}_2) + K_1 x_1 + K_2(x_1 - x_2)$$

$$F_1(s) = M_1 s^2 X_1(s) + B_1 s(X_1(s) - X_2(s)) + K_1 X_1(s) + K_2(X_1(s) - X_2(s))$$

$$\boxed{F_1(s) = X_1(s)\left[M_1 s^2 + B_1 s + (K_1 + K_2)\right] - X_2(s)\left[B_1 s + K_2\right]} \rightarrow (1).$$

free body diagram for M2.



(No external forces)

$f_{M_2} = M_2 \ddot{x}_2$

$f_{B_1} = B_1(\dot{x}_2 - \dot{x}_1)$

$f_{B_2} = B_2 \dot{x}_2$

$f_{K_2} = K_2(x_2 - x_1)$

$$0 = M_2\ddot{x}_2 + B_1(\dot{x}_2 - \dot{x}_1) + B_2\dot{x}_2 + K_2(x_2 - x_1).$$

$$0 = M_2 s^2 X_2(s) + B_1 s(X_2(s) - X_1(s)) + B_2 X_2(s) s + K_2(X_2(s) - X_1(s)).$$

$$0 = X_2(s)\left(M_2 s^2 + (B_1 + B_2)s\right) + K_2 X_2(s)) - (B_1 s + K_2)X_1(s)$$

$$\boxed{X_1(s) = \frac{[M_2 s^2 + (B_1+B_2)s + K_2]\, X_2(s)}{B_1 s + K_2}.}$$

Subtitute $X_1(s)$ in eqn ①.

$$F_1(s) = \frac{[M_2 s^2 + (B_1+B_2)s + K_2]}{B_1 s + K_2} X_2(s) [M_1 s^2 + B_1 s + (K_1+K_2)] - [X_2(s)[B_1 s + K_2]]$$

$$\frac{F_1(s)}{X_2(s)} = \frac{[M_2 s^2 + (B_1+B_2)s + K_2][M_1 s^2 + B_1 s + (K_1+K_2)] - [B_1 s + K_2]^2}{B_1 s + K_2}$$

$$\frac{X_2(s)}{F_1(s)} = \frac{B_1 s + K_2}{K_1K_2 + \underline{b_1 m_1 s^3 + b_1 m_2 s^3 + b_2 m_1 s^3} + \underline{k_1 m_2 s^2 + k_2 m_1 s^2 + k_2 m_2 s^2} + \underline{m_1 m_2 s^4} + \underline{b_1 k_1 s + b_2 k_1 s + b_2 k_2 s} + \underline{b_1 b_2 s^2}}$$

$$\frac{X_2(s)}{F_1(s)} = \frac{B_1 s + K_2}{m_1 m_2 s^4 + (b_1 m_1 + b_1 m_2 + b_2 m_1)s^3 + (k_1 m_2 + k_2 m_1 + k_2 m_2 + b_1 b_2)s^2 + (b_1 k_1 + b_2 k_1 + b_2 k_2)s + K_1 K_2}$$

Here degree of $D(s) = 4$.

$\boxed{\text{hint}}$ use matlab Symbolic tool box.

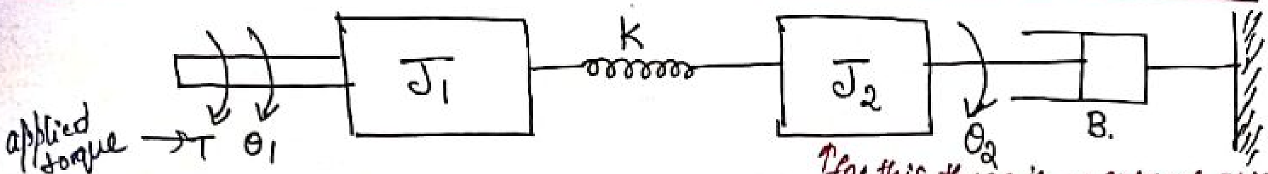
Syms M1, Syms m2;

:

expand (tf)

Pretty (tf).

Find the transfer function of mechanical rotational system.



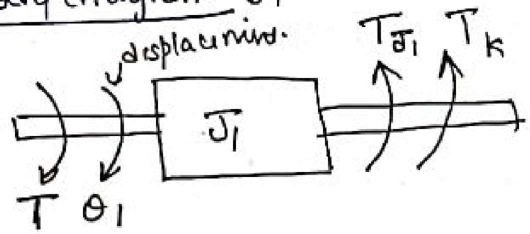
↑ for this there is no external applied torque

for J_1 & J_2 we should draw free body diagram.

Note: Angular displacement, Angular velocity, Angular acceleration

$$T.F \Rightarrow \frac{\theta_2(s)}{\cancel{\theta_1}T(s)}$$

Free body diagram J_1



$$T_{J_1} = J_1 \cdot \ddot{\theta}$$

$$T_K = K \cdot (\theta_1 - \theta_2)$$

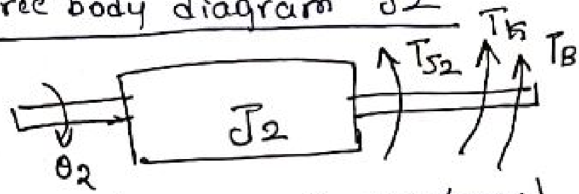
$$T = T_{J_1} + T_K.$$

$$T = J_1 \ddot{\theta}_1 + K(\theta_1 - \theta_2)$$

$$T(s) = J_1 s^2 \theta_1(s) + K\theta_1(s) - K\theta_2(s)$$

$$\boxed{T(s) = (J_1 s^2 + K)\,\theta_1(s) + K \cdot \theta_2(s)} \rightarrow (1)$$

Free body diagram J_2



(applied torque = opposing torque)

$$0 = T_{J_2} + T_K + T_B$$

$$0 = J_2 \ddot{\theta}_2 + K(\theta_2 - \theta_1) + B \cdot \dot{\theta}_2$$

take LT

$$0 = J_2 s^2 \theta_2(s) + B \cdot s\, \theta_2(s) + K(\theta_2(s) - \theta_1(s))$$

$$\theta_2(s)\left[J_2 s^2 + Bs + K\right] = K \cdot \theta_1(s)$$

$$\boxed{\theta_1(s) = \theta_2(s)\frac{(J_2 s^2 + Bs + K)}{K}} \rightarrow (2)$$

Substituting (2) in (1)

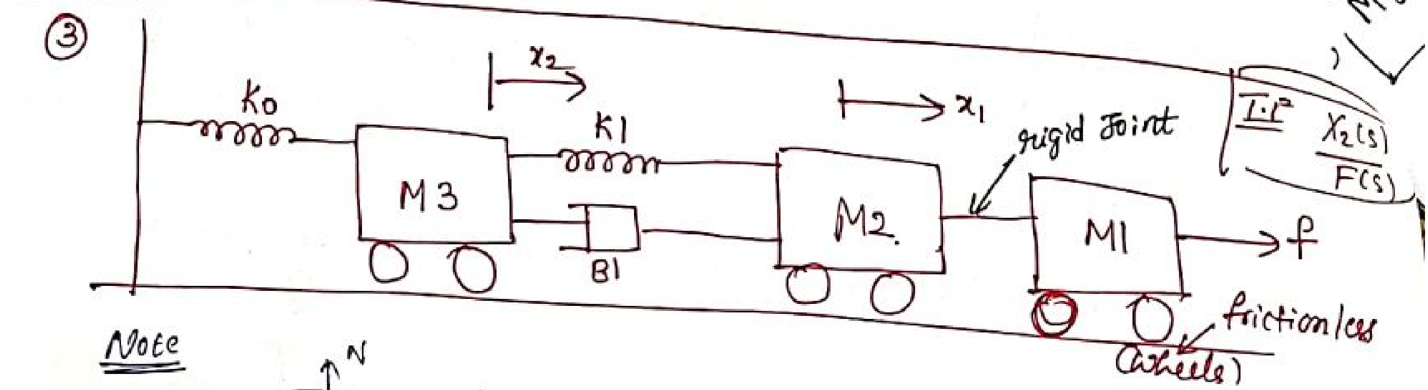
$$T(s) = (J_1 s^2 + K)\frac{(J_2 s^2 + Bs + K)}{K}\,\theta_2(s) + K \cdot \theta_2(s)$$

$$\frac{T(s)}{\theta_2(s)} = \frac{(J_1 s^2 + K)(J_2 s^2 + Bs + K) + K^2}{K}$$

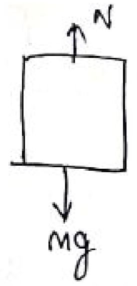
Slope at $x = x_0$ ... terms

$$\frac{\theta_2(s)}{T(s)} = \frac{K}{(J_1 s^2+K)(J_2 s^2+Bs+K)+K^2}$$

③



Note



Note when M is moving in horizontal direction, N & Mg forces are not there

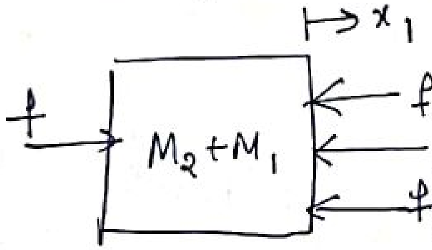
Greater M ⇒ Greater Inertia.

Soln

no of displacements = no of independent mass.

(M_1 & M_2) are connected through rigid Joint, M_1 & M_2 are not independent.

free body diagram for M_2+M_1



$f_{M_1+M_2} = (M_2+M_1)\ddot{x}_1$

$f_{K_1} = (x_1-x_2)\cdot K_1$

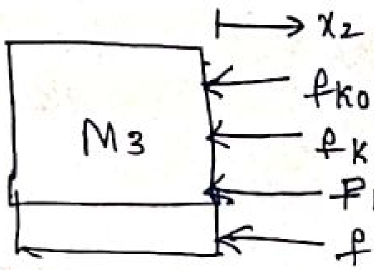
$f_{B_1} = (\dot{x}_1-\dot{x}_2)B_1$

$$(M_2+M_1)\ddot{x}_1 + (\dot{x}_1-\dot{x}_2)B_1 + (x_1-x_2)K_1 = f$$

$$\boxed{(M_2+M_1)s^2X_1(s) + B_1 s(X_1(s)-X_2(s)) + K_1(X_1(s)-X_2(s)) = F(s)} \rightarrow ①$$

free body diagram for M3

(No external force)



$f_{K_0} = K_0 x_2$

$f_{K_1} = K_1(x_2-x_1)$

$f_B = B_1(\dot{x}_2-\dot{x}_1)$

$f_{M_3} = M_3\cdot\ddot{x}_2$

$$\boxed{M_3 s^2 X_2(s) + k_0 X_2(s) + k_1 (X_2(s) - X_1(s)) + B_1 s (X_2(s) - X_1(s)) = 0} \rightarrow (2)$$

from Eq^n (2)

$$(X_1(s) - X_2(s))(k_1 + B_1 s) = (M_3 s^2 + k_0) X_2(s)$$

$$\boxed{X_1(s) - X_2(s) = \frac{M_3 s^2 + k_0}{k_1 + B_1 s} X_2(s)} \rightarrow (3)$$

from Eq^n (2).

$$(M_3 s^2 + B_1 s + (k_1 + k_0)) X_2(s) = (k_1 + B_1 s) X_1(s).$$

or $$\boxed{X_1(s) = \frac{M_3 s^2 + Bs + (k_1 + k_0)}{k_1 + B_1 s} X_2(s).} \rightarrow (4).$$

Substitute (3) & (4) in (1)

$$X_2(s) (M_2 + M_1) s^2 \frac{(M_3 s^2 + Bs + (k_1 + k_0))}{(k_1 + B_1 s)} + (B_1 s + k_1)\left(\frac{M_3 s^2 + k_0}{k_1 + B_1 s}\right) X_2(s) = F(s)$$

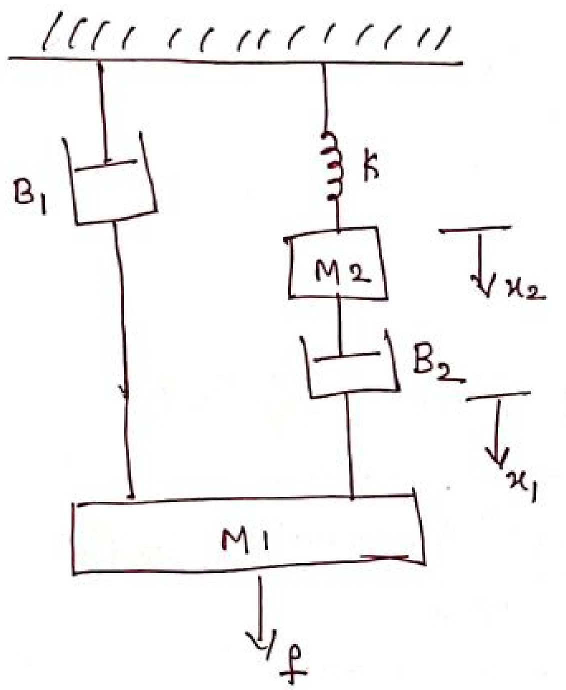
$$\boxed{\frac{X_2(s)}{F(s)} = \frac{k_1 + B_1 s}{(M_2 + M_1) s^2 (M_3 s^2 + Bs + (k_1 + k_0)) + (M_3 s^2 + k_0)(k_1 + B_1 s)}}$$

Order of the System = 4. (No of energy storing elements $M_3, M_2 + M_1, k_1, k_0$)

No of zero's
- finite = 1
- at infinity = 3.

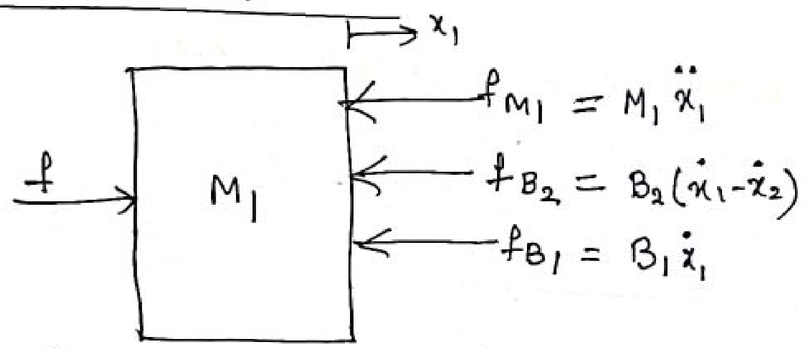
slope at $x = x_0$ ... terms.

④



find TF $\dfrac{X_2(s)}{F(s)}$

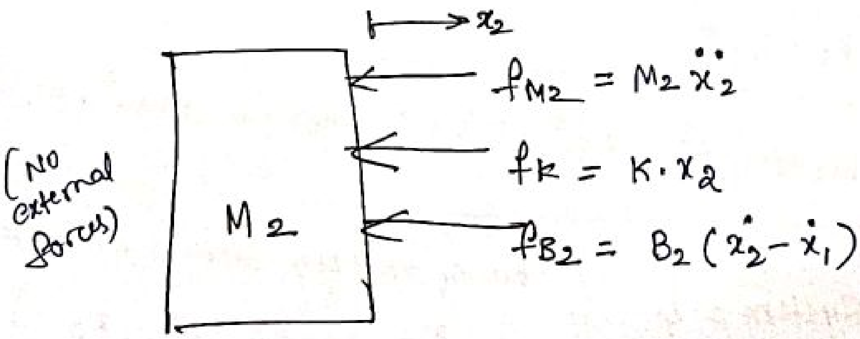
free Body diagram for M1



$$M_1 \ddot{x}_1 + B_1 \dot{x}_1 + B_2 (\dot{x}_1 - \dot{x}_2) = f$$

$$\boxed{M_1 s^2 X_1(s) + B_1 s X_1(s) + B_2 s (X_1(s) - X_2(s)) = F(s)} \rightarrow (1)$$

free body diagram for M2



$$M_2 \ddot{x}_2 + K x_2 + B_2 (\dot{x}_2 - \dot{x}_1) = 0$$

$$\boxed{M_2 s^2 X_2(s) + K X_2(s) + B_2 s (X_2(s) - X_1(s)) = 0} \rightarrow (2)$$

from 2

$$B_2 s\left(X_1(s) - X_2(s)\right) = \left(M_2 s^2 + k\right) X_2(s) \longrightarrow ③$$

$$X_1(s) = \left(\frac{M_2 s^2 + B_2 s + k_1}{B_2 s}\right) X_2(s) \longrightarrow ④$$

sub ③ & ④ in ①

$$(M_1 s^2 + B_1 s) \cdot \frac{(M_2 s^2 + B_2 s + k_1)}{B_2 s} X_2(s) + (B_2 s) \frac{(M_2 s^2 + k) X_2(s)}{B_2 s} = F(s)$$

$$\frac{X_2(s)}{F(s)} = \frac{B_2 \cancel{(s)}}{\cancel{s}\left[(M_1 s + B_1)(M_2 s^2 + B_2 s + k_1) + B_2(M_2 s^2 + k)\right]}$$

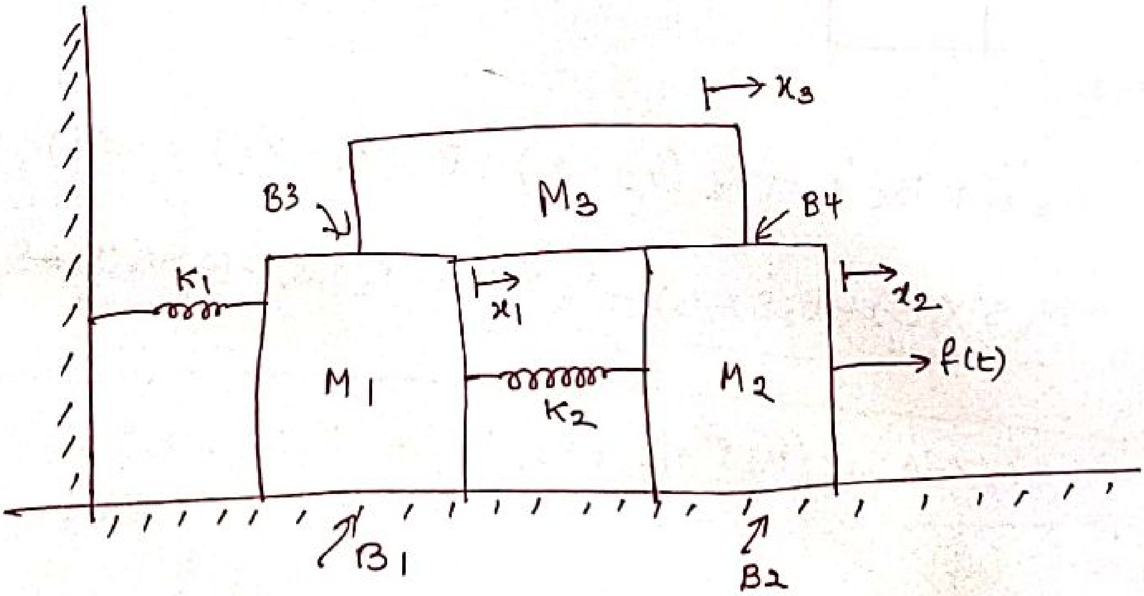
order of system = 3.

no of zeros ⇒ finite (at origin)

at infinity - 3

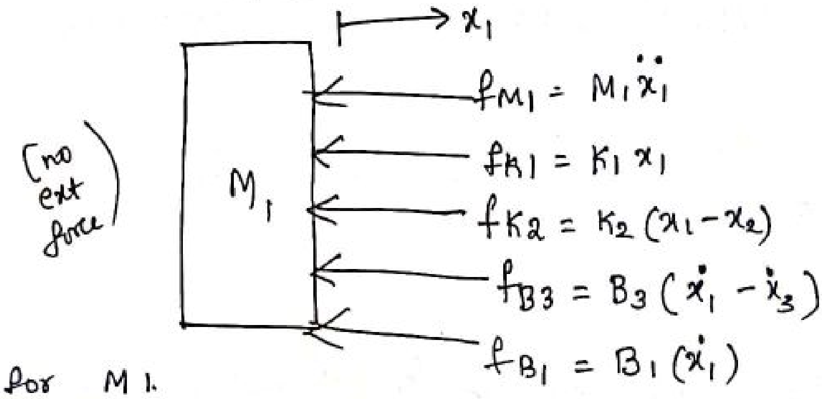
No. of energy storage elements = 3 (M_1, M_2, k)

5



Slope at $x = x_0$ HOT ⇒ Higher order terms.

free body diagram M1



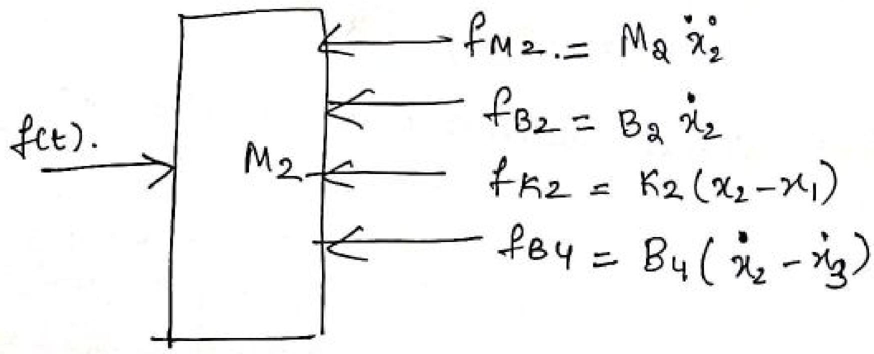
for M1.

$$B_1 \dot{x}_1 + M_1 \ddot{x}_1 + K_1 x_1 + B_3 (\dot{x}_1 - \dot{x}_3) + K_2 (x_1 - x_2) = 0$$

$$B_1 s X_1(s) + M_1 s^2 X_1(s) + K_1 X_1(s) + B_3 s (X_1(s) - X_3(s)) + K_2 (X_1(s) - X_2(s)) = 0$$

$$\boxed{(M_1 s^2 + (B_1 + B_3) s + (K_1 + K_2)) X_1(s) - K_2 X_2(s) - B_3 s X_3(s) = 0} \rightarrow (1)$$

free body diagram M2



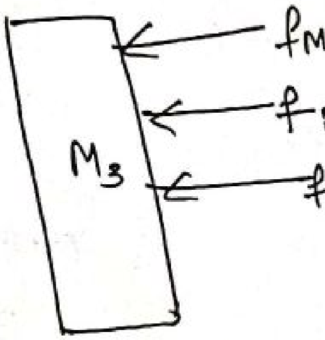
for M2

$$M_2 \ddot{x}_2 + B_2 \dot{x}_2 + K_2 (x_2 - x_1) + B_4 (\dot{x}_2 - \dot{x}_3) = f(t).$$

$$M_2 s^2 X_2(s) + B_2 s X_2(s) + K_2 X_2(s) - K_2 X_1(s) + B_4 s X_2(s) - B_4 s X_3(s) = F(s)$$

$$\boxed{-K_2 X_1(s) + (M_2 s^2 + (B_2 + B_4) s + K_2) X_2(s) - B_4 s X_3(s) = F(s)} \rightarrow (2)$$

free body diagram M3.



$f_{M3} = M_3 \ddot{x}_3$

$f_{B3} = B_3(\dot{x}_3 - \dot{x}_1)$

$f_{B4} = B_4(\dot{x}_3 - \dot{x}_2)$

(no ext force)

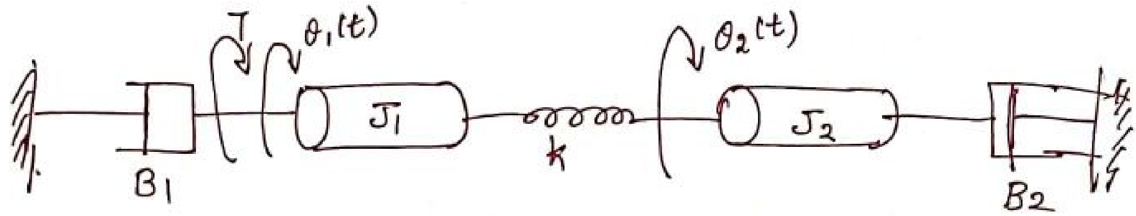
$$M_3 \ddot{x}_3 + B_3(\dot{x}_3 - \dot{x}_1) + B_4(\dot{x}_3 - \dot{x}_2) = 0$$

$$M_3 s^2 X_3(s) + B_3 s \cdot X_3(s) - B_3 s \cdot X_1(s) + B_4 s \cdot X_3(s) - B_4 s \cdot X_2(s) = 0$$

$$\boxed{-B_3 s X_1(s) - B_4 s \cdot X_2(s) + \left(M_3 s^2 + (B_4 + B_3)s\right) X_3(s) = 0} \rightarrow (3)$$

find. $F(s)$

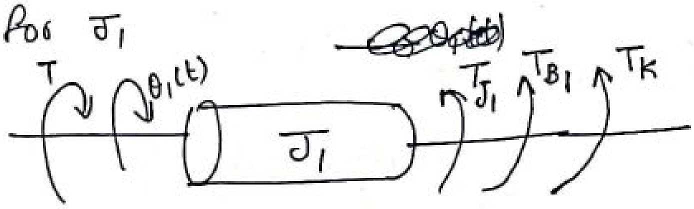
⑥



find $\dfrac{\theta_1(S)}{T(S)}$ & $\dfrac{\theta_2(S)}{T(S)}$

Note
θ : function of time
$\theta(t)$
$\dfrac{d\theta(t)}{dt} = \dot{\theta}(t)$
$\dfrac{d^2\theta(t)}{dt^2} = \ddot{\theta}(t)$

For J_1



$$J_1\ddot{\theta}_1 + B_1\dot{\theta}_1 + K(\theta_1 - \theta_2) = T$$

$$\boxed{J_1 s^2\theta_1(S) + B_1 S\theta_1(S) + K(\theta_1(S) - \theta_2(S)) = T(S)} \rightarrow ①$$

$$J_2\ddot{\theta}_2 + B_2\cdot\dot{\theta}_2 + K(\theta_2 - \theta_1) = 0$$

$$\boxed{J_2 s^2\theta_2(S) + B_2 S\,\theta_2(S) + K(\theta_2(S) - \theta_1(S)) = 0} \rightarrow ②$$

from ②

$$\cancel{K}\,(\theta_1(S) - \theta_2(S)) = \frac{(J_2 s^2 + B_2 S)\theta_2(S)}{K} \longrightarrow ③$$

$$\theta_1(S) = \frac{(J_2 s^2 + B_2 S + K)}{K}\theta_2(S) \longrightarrow ④$$

substituting ③ & ④ in ①

$$(J_1 s^2 + B_1 S)\frac{(J_2 s^2 + B_2 S + K)}{K}\theta_2(S) + K.\frac{(J_2 s^2 + B_2 S)\theta_2(S)}{K} = T(S)$$

$$\frac{\theta_2(S)}{T(S)} = \frac{K}{(J_1 s^2 + B_1 S)(J_2 s^2 + B_2 S + K) + K(J_2 s^2 + B_2 S)} \rightarrow ⑤$$

(7)

from (1)

$$(J_1 s^2 + B_1 s + K)\,\theta_1(s) - K\cdot\theta_2(s) = T(s)$$

$$(J_1 s^2 + B_1 s + K)\,\theta_1(s) - K\cdot\frac{K}{J_2 s^2 + B_2 s + K}\cdot\theta_1(s) = T(s)$$

$$\frac{\left((J_1 s^2 + B_1 s + K)(J_2 s^2 + B_2 s + K) - K^2\right)\theta_1(s)}{J_2 s^2 + B_2 s + K} = T(s).$$

$$\theta_2(s) = \frac{K\cdot\theta_1(s)}{J_2 s^2 + B_2 s + K.}$$

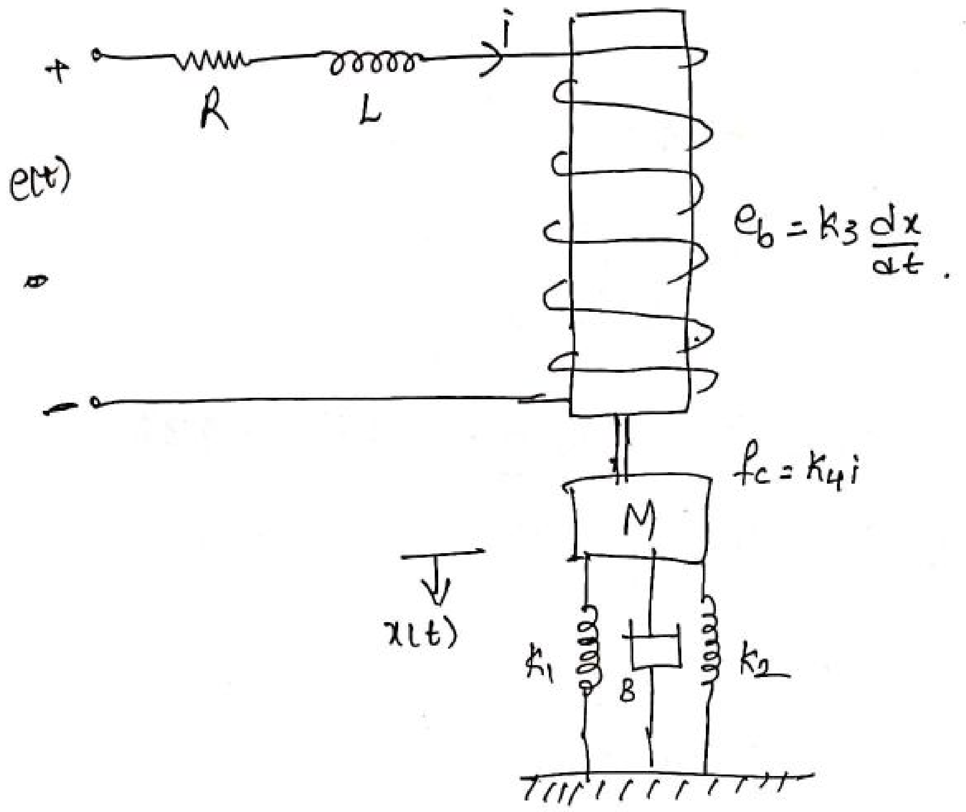
Substituting in (5).

$$\frac{\cancel{K}\cdot\theta_1(s)}{T(s)\cdot(J_2 s^2 + B_2 s + K)} = \frac{K}{(J_1 s^2 + B_1 s)(J_2 s^2 + B_2 s + K) + (J_2 s^2 + B_1 s)K}$$

$$\boxed{\frac{\theta_1(s)}{T(s)} = \frac{J_2 s^2 + B_2 s + K.}{(J_1 s^2 + B_1 s)(J_2 s^2 + B_2 s + K) + (J_2 s^2 + B_1 s)K.}}$$

---

(7) consider the electro-mechanical system. Determine the TF $X(s)/E(s)$. The coil has back emf $e_b = K_3 \frac{dx}{dt}$ & the coil current i produces force $f_c = K_4\, i$ on mass M



consider the electrical network

$$e(t) = R\,i(t) + L \cdot \frac{di(t)}{dt} + e_b(t)$$

$$e(t) = R\,i(t) + L \cdot \frac{di(t)}{dt} + K_3 \frac{dx}{dt} \longrightarrow (1)$$

consider the mechanical network.

$$f_c = M \cdot \frac{d^2x}{dt} + B \cdot \frac{dx}{dt} + (K_1 + K_2)\,x = K_4 i \rightarrow (2)$$

under zero initial conditions

$$E(S) = R\,I(S) + LS\,I(S) + K_3 S \cdot X(S) \longrightarrow (3).$$

$$F_c(S) = MS^2 X(S) + BS \cdot X(S) + (K_1 + K_2)\,X(S)$$

$$K_4 I(S) = (MS^2 + BS + (K_1 + K_2))\,X(S) \longrightarrow (4)$$

from (3) $I(S) = \dfrac{E(S) - K_3 S \cdot X(S)}{R + LS} \rightarrow (5)$

1

from ⑤ & ④

$$K_4 \cdot \left(\frac{E(s) - K_3 s X(s)}{R + Ls}\right) = (Ms^2 + Bs + (K_1 + K_2)) . X(s)$$

$$\frac{K_4 \cdot E(s)}{R + Ls} = \left[(Ms^2 + Bs + (K_1 + K_2)) + \frac{K_3 K_4 \cdot s}{R + Ls} \right] X(s)$$

$$\boxed{\frac{X(s)}{E(s)} = \frac{K_4}{(Ms^2 + Bs + (K_1 + K_2))(R + Ls) + K_3 K_4 s}.}$$

order = 3

no of zero at infinity = 3.

no of storage elements = 3 = M, $(K_1 + K_2)$, L.

terms.

Linear Approximations of Physical Systems.

Most non-linear systems can be approximated as linear system about an operating point.

Consider the system $\boxed{y = mx + b}$ or $\boxed{y = x^2}$, is nonlinear, because superposition theorem is not satisified.

Let us take $y = mx + b$ about an operating point (x_0, y_0) for a small changes Δx & Δy.

$$y = mx + b.$$

$$y_0 + \Delta y = m x_0 + m\Delta x + b.$$

Therefore $\Delta y = m\Delta x$, satisfies linearity conditions.

In general, physical systems are non linear, but is linear within some range of variable.

Linearising a non-linear process

① Expand in Taylor series about the operating point or equilibrium point

② Evaluate the first derivative about the operating / equilibrium point to get the linear approximation.

The relationship of two variables is written as

$y(t) = g(x(t))$ where $y(t)$ is non linear function of $x(t)$.

The Taylor series expansion about the operating point x_0

$$y = g(x) = g(x_0) + \underbrace{\frac{dg}{dx}}_{\text{Slope at } x = x_0}\bigg|_{x=x_0} \frac{x - x_0}{1!} + \underbrace{\frac{d^2g}{dx^2}\bigg|_{x=x_0} \frac{(x-x_0)^2}{2!} + \cdots}_{\text{HOT} \Rightarrow \text{Higher order terms.}}$$

Neglecting HOT

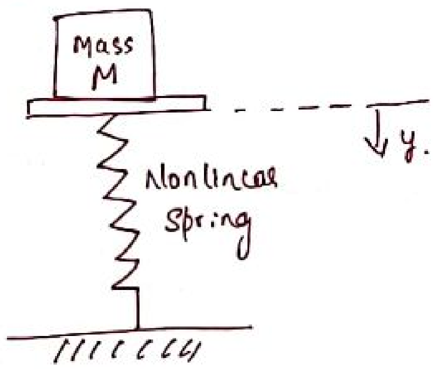
$$y = \underbrace{g(x_0)}_{y_0} + \underbrace{\left.\frac{dg}{dx}\right|_{x=x_0}}_{m} (x - x_0)$$

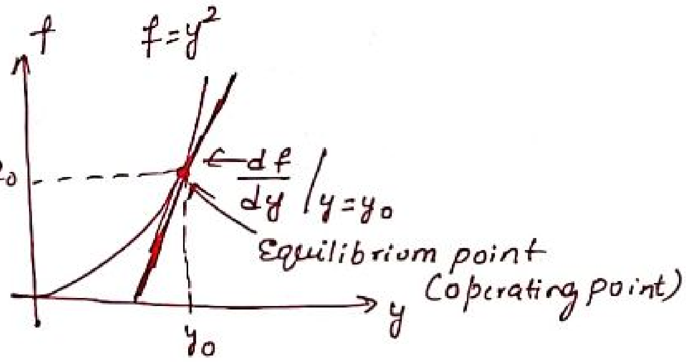
$$y - y_0 = m\,(x - x_0)$$

$$\boxed{\Delta y = m \cdot \Delta x} \rightarrow \text{linear Equation}$$

So all the non-linear elements are approximated as linear.

Eg. ① Non linear spring





- consider mass M sitting on non-linear spring.
- Normal operating point is the equilibrium position that occurs when spring force balances gravitational force Mg, where g = gravitational constant. Mass

$$\therefore \boxed{f_0 = Mg}$$

For non-linear spring $f = y^2$

or $y_0 = \sqrt{f_0} = \sqrt{Mg}$.

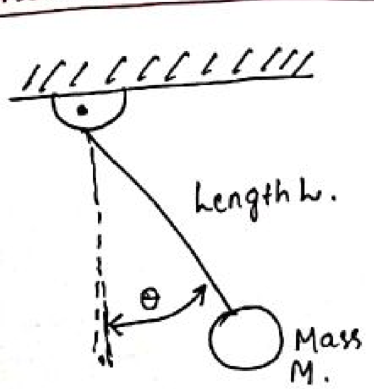
$\therefore$ The linear model for small deviation is

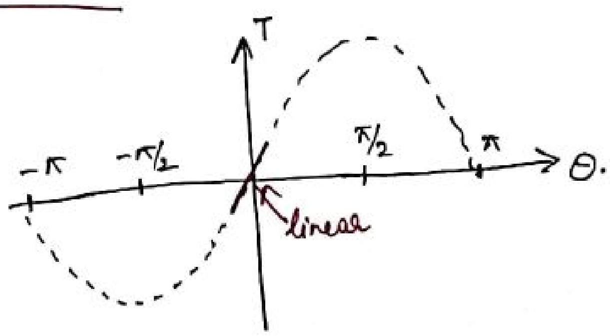
$$\boxed{\Delta f = \underbrace{m}_{\text{Slope}} \Delta y} \quad \text{where } m = \left.\frac{df}{dy}\right|_{y=y_0}.$$

$$\therefore m = 2y_0.$$

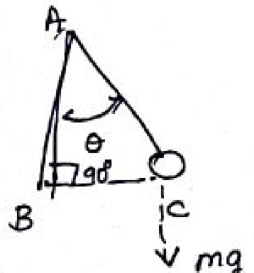
Eg 2:

Pendulum Oscillator Model





Consider the pendulum shown in figure. The torque on the mass is



$\sin\theta = \frac{BC}{AC} = \frac{BC}{L}$ or $BC = L\sin\theta$

$L \Rightarrow$ length of spring.

$$\boxed{T = mgL\sin\theta} \rightarrow (1) \Rightarrow \text{non linear.}$$

Gravitational torque

→ Equilibrium state/condition for the mass $\theta_0 = 0$

1st derivative evaluated at equilibrium

$$T = T_0 + mgL \cdot \frac{\partial \sin\theta}{\partial\theta}\bigg|_{\theta=\theta_0} (\theta - \theta_0)$$

$$T = T_0 + mgL\cos\theta\bigg|_{\theta=\theta_0} (\theta - \theta_0)$$

Since $\theta_0 = 0$ $\quad T_0 = mgL\sin\theta_0 = 0$

$$T = mgL \overset{1}{\cancel{\cos(0)}} (\theta - 0)$$

$$\boxed{T = mgL\theta} \text{ (2) linear.}$$

This approximation is reasonably accurate for.

$$-\frac{\pi}{4} < \theta < \frac{\pi}{4}.$$

Note:

For many variables.

$$y = g(x_1, x_2, \ldots x_n)$$

Neglecting the higher order terms, the linear approximation would be.

Taylor series expansion about operating point $x_{10}, x_{20} \ldots x_{n0}$

$$y = g(x_{10}, x_{20}, \ldots x_{n0}) + \left.\frac{\partial g}{\partial x_1}\right|_{x=x_0} (x_1 - x_{10}) + \left.\frac{\partial g}{\partial x_2}\right|_{x=x_0} (x_2 - x_{20})$$

$$+ \cdots + \left.\frac{\partial g}{\partial x_n}\right|_{x=x_0} (x_n - x_{n0})$$

where x_0 = operating point.

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